

计算机视觉 ——目标识别

2026年春季

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目的

- 了解什么是目标识别？
- 熟悉两种目标识别方法



作业

- 分析HOG、颜色直方图、PCA、shapecontext、特征点+BOW这几种表达的平移、旋转（平面内）、尺度、遮挡不变性，并分析原因

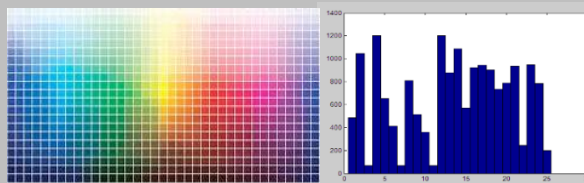
	平移	旋转	尺度	遮挡
HOG	✗	✗	✗	✗
颜色直方图	✓?	✓	✓?	✗
PCA	✓	✓	✓?	✗
shapecontext	✓	✓	✓	✓?
特征点+BOW	✓	✓?	✓?	✗

图像特征

Input image



Color: Quantize RGB values



Invariance?

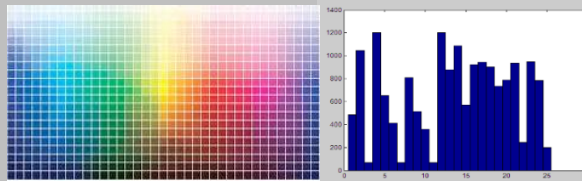
- ? Translation
- ? Scale
- ? Rotation
- ? Occlusion

图像特征

Input image



Color: Quantize RGB values



Invariance?

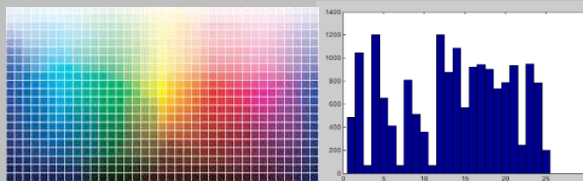
- 😊 Translation
- ? Scale
- 😊 Rotation (in-planar)
- 😞 Occlusion

图像特征

Input image



Color: Quantize RGB values



Invariance?

- 😊 Translation
- 😊 Scale
- 😊 Rotation (in-planar)
- 😞 Occlusion

Global shape: PCA space



Invariance?

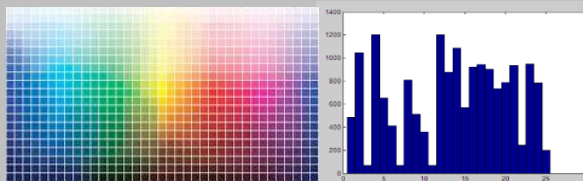
- ? Translation
- ? Scale
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图像特征

Input image



Color: Quantize RGB values



Invariance?

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Global shape: PCA space



Invariance?

- 😊 Translation
- ? Scale
- 😊 Rotation (in-planar)
- 😞 Occlusion

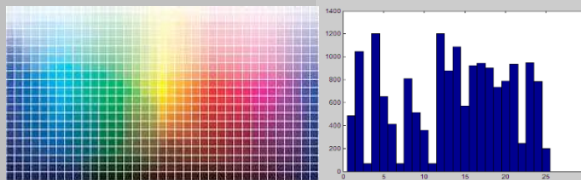


图像特征

Input image



Color: Quantize RGB values



Invariance?

- 😊 Translation
- 😊 Scale
- 😊 Rotation
- 😞 Occlusion

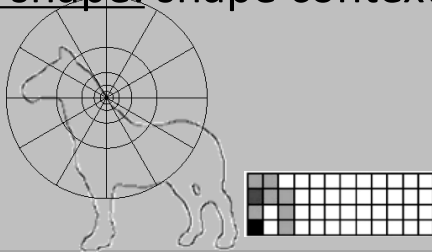
Global shape: PCA space



Invariance?

- 😊 Translation
- ? Scale
- 😊 Rotation
- 😞 Occlusion

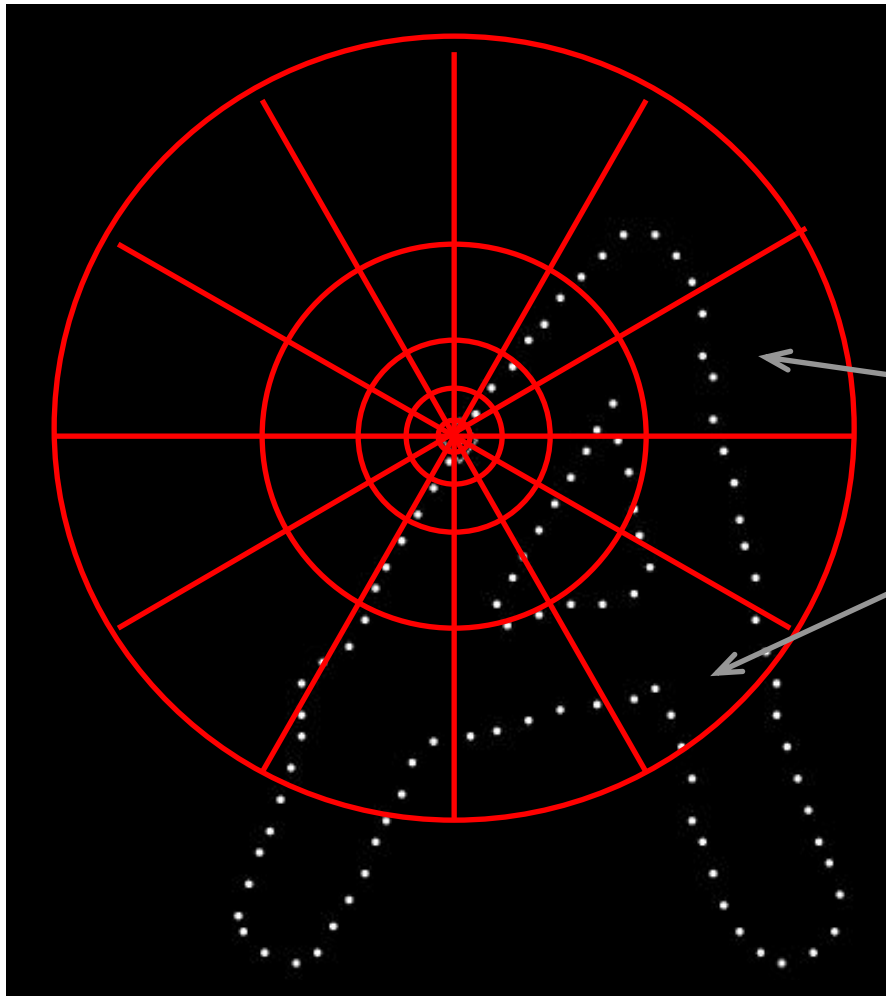
Local shape: shape context



Invariance?

- ? Translation
- ? Scale
- ? Rotation (in-planar)
- ? Occlusion

Local Descriptors: Shape Context



Count the number of points
inside each bin, e.g.:

Count = 4

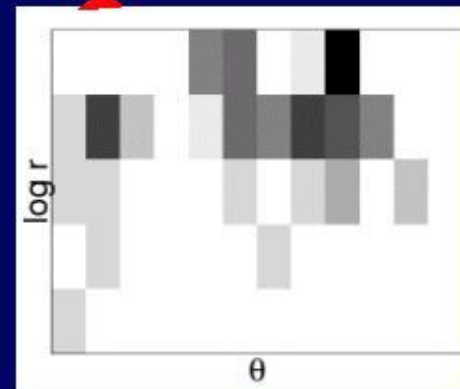
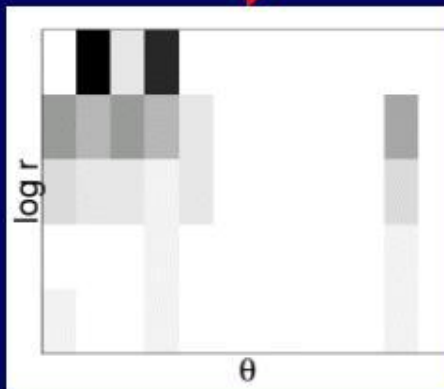
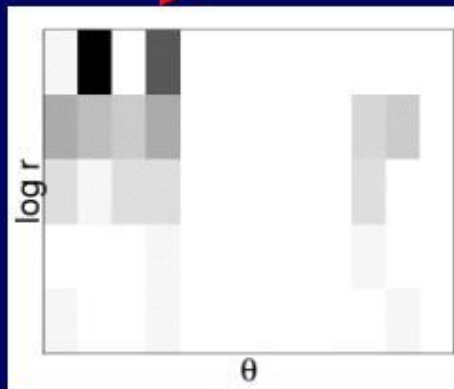
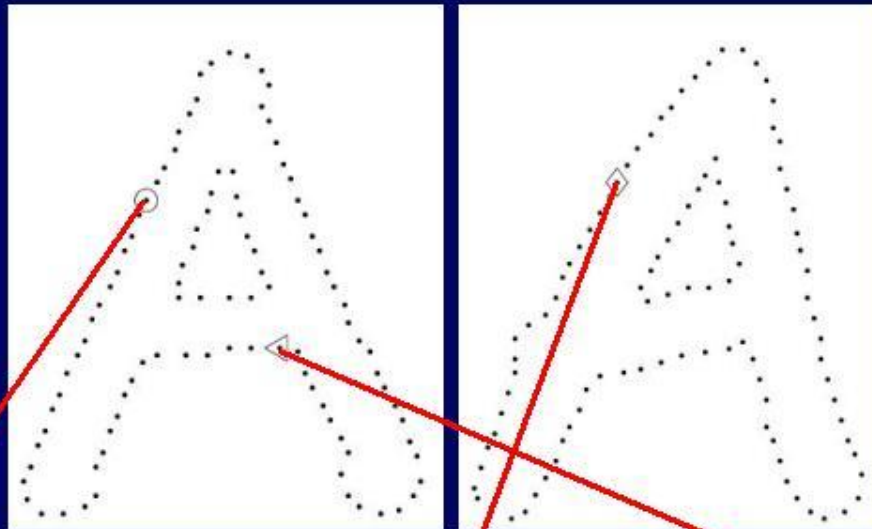
⋮

Count = 10

Log-polar binning: more
precision for nearby points,
more flexibility for farther
points.



Shape Context Descriptor



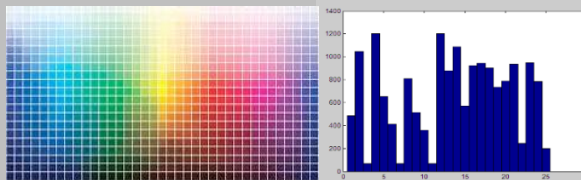


图像特征

Input image



Color: Quantize RGB values



Invariance?

- 😊 Translation
- 😊 Scale
- 😊 Rotation
- 😞 Occlusion

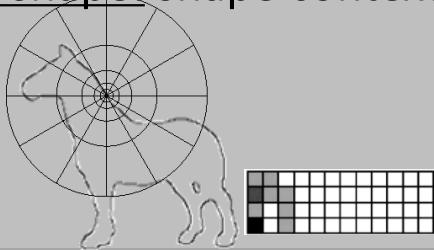
Global shape: PCA space



Invariance?

- 😊 Translation
- ? Scale
- 😊 Rotation
- 😞 Occlusion

Local shape: shape context



Invariance?

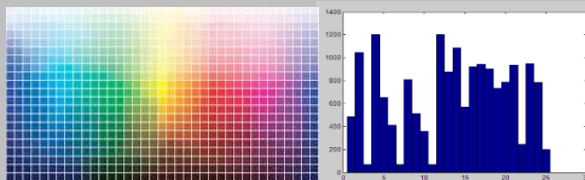
- 😊 Translation
- 😊 Scale
- ? Rotation (in-planar)
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图像特征

Input image



Color: Quantize RGB values



Invariance?

- 😊 Translation
- 😊 Scale
- 😊 Rotation
- 😞 Occlusion

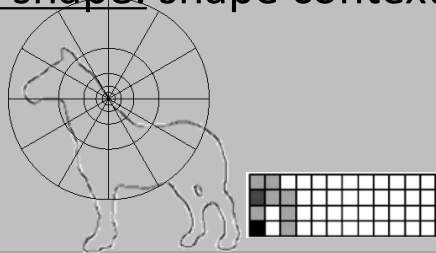
Global shape: PCA space



Invariance?

- 😊 Translation
- ? Scale
- 😊 Rotation
- 😞 Occlusion

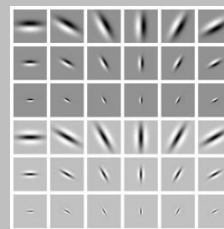
Local shape: shape context



Invariance?

- 😊 Translation
- 😊 Scale
- ? Rotation (in-planar)
- 😞 Occlusion

Texture: Filter banks



Invariance?

- ? Translation
- ? Scale
- ? Rotation (in-planar)
- ? Occlusion

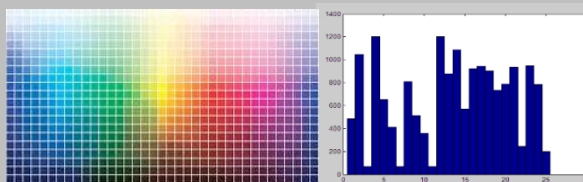


图像特征

Input image



Color: Quantize RGB values



Invariance?

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- 😊 Scale
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- 😞 Occlusion

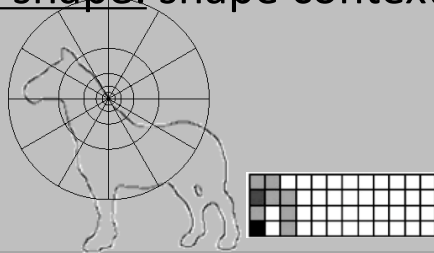
Global shape: PCA space



Invariance?

- 😊 Translation
- ? Scale
- 😊 Rotation
- 😞 Occlusion

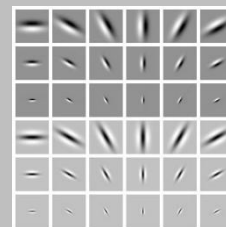
Local shape: shape context



Invariance?

- 😊 Translation
- 😊 Scale
- ? Rotation (in-planar)
- 😞 Occlusion

Texture: Filter banks



Invariance?

- 😊 Translation
- ? Scale
- ? Rotation (in-planar)
- 😞 Occlusion



目标识别（分类）

- 任务：通用目标识别





目标识别（分类）

- 任务：通用目标识别



horse



person



目标识别（分类）

- 基于特征点的目标识别



- 局部特征 + BoW



Bag-of-words 目标识别框架

Object



Bag of 'words'



(Li Fei-Fei, ICCV 2005)



Bag-of-words 目标识别框架

Of all the sensory impressions proceeding to the brain, the visual experiences are the dominant ones. Our perception of the world around us is based on the messages that our eyes. For a long time, sensory, brain, visual, perception, retinal, cerebral cortex, eye, cell, optical nerve, image Hubel, Wiesel

image falling on the retina undergoes a step-wise analysis in a system of nerve cells organized in columns. In this system each cell has a specific function and is responsible for a specific detail in the pattern of the retinal image.

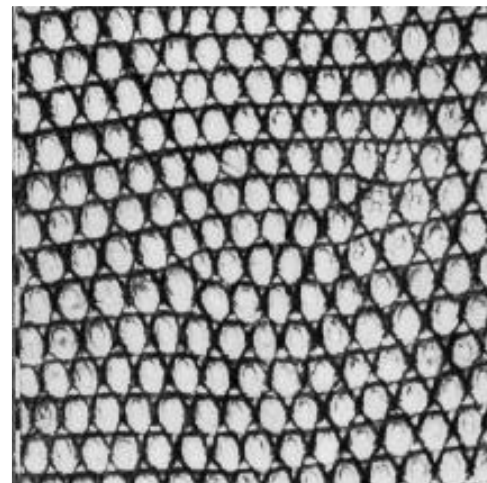
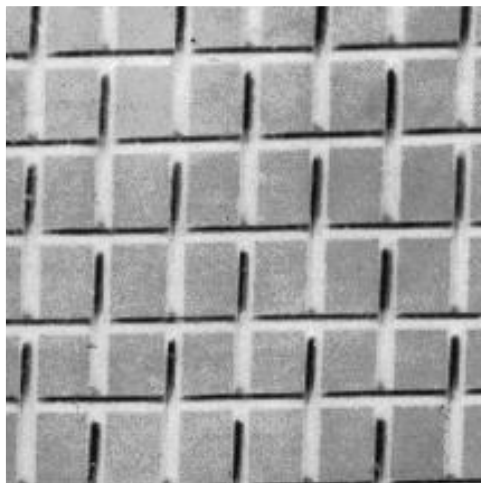
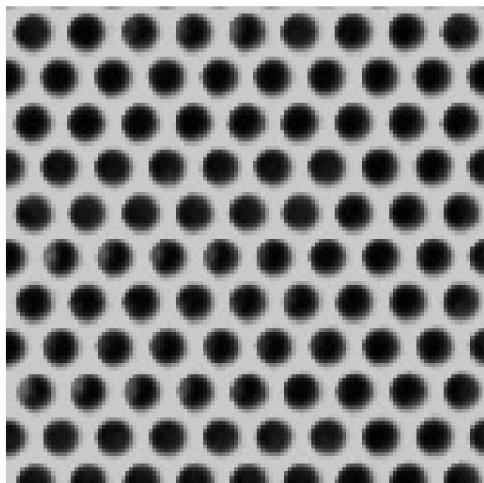
China is forecasting a trade surplus of \$90bn (£51bn) to \$100bn this year, a threefold increase on 2004's \$32bn. The Commerce Ministry said the surplus could be created by a predicted 30% increase in exports to \$750bn, compared with \$560bn. This will annoy the US. China's deliberate policy to increase exports of yuan is a government policy. The government also needs to increase demand for yuan in the country. China's trade surplus with the US against the yuan is a government policy and permitted it to trade freely. However, Beijing has not clear that it will take its time and trade carefully before allowing the yuan to increase further in value.

China, trade, surplus, commerce, exports, imports, US, yuan, bank, domestic, foreign, increase, trade, value



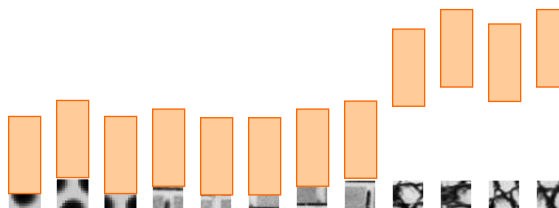
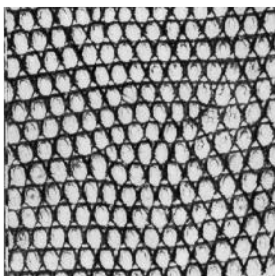
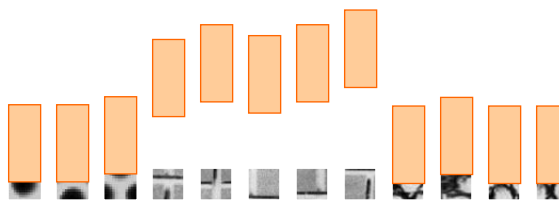
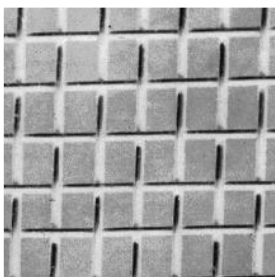
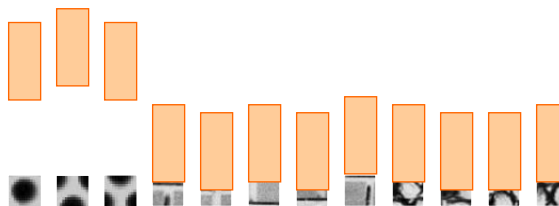
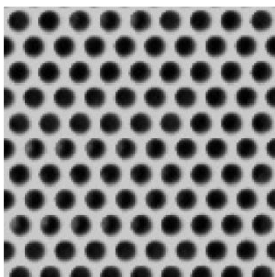
Bag-of-words 目标识别框架

- 纹理是由一些重复的基元（texton）组成的
- 对于有明显统计特征的纹理来讲，我们只需要确认其基元，而不需要确定空间排列





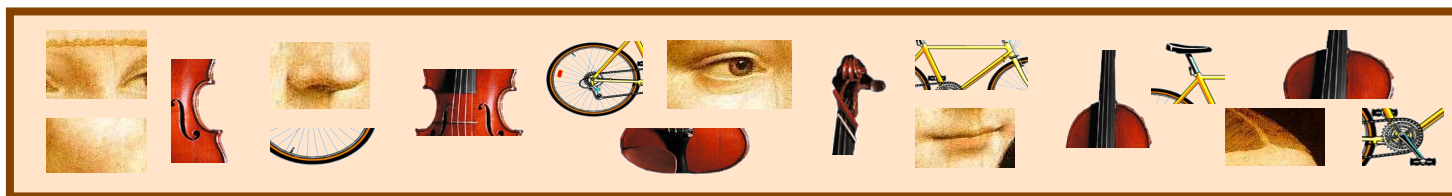
Bag-of-words 目标识别框架





Bag-of-words 目标识别框架

- Bag-of-words 框架

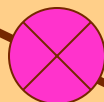


$$P(w_i|d_j) = \sum_{k=1}^K P(z_k|d_j)P(w_i|z_k),$$

learning



feature detection
& representation



codewords dictionary

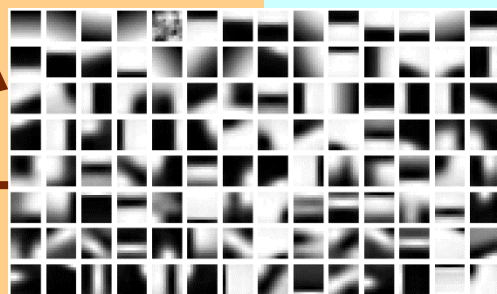
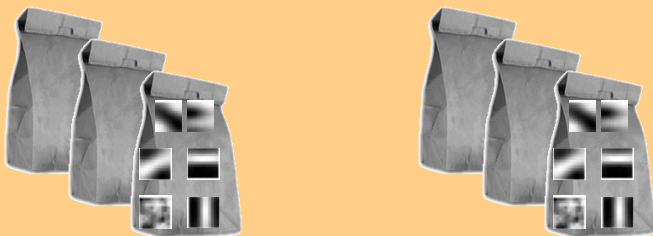
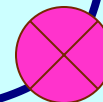


image representation



**category models
(and/or) classifiers**

recognition

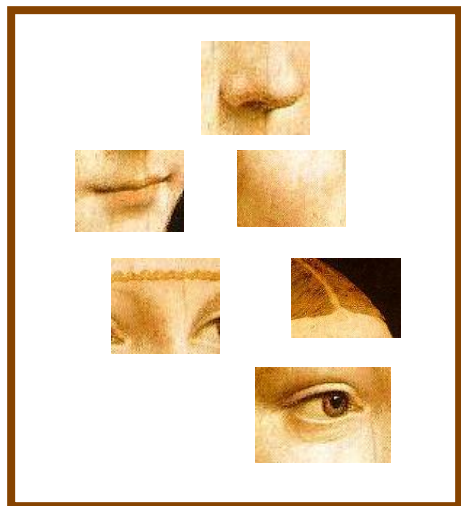


**category
decision**



基于Bags of features的目标识别

1. 特征提取





基于Bags of features的目标识别

1. 特征提取
2. 学习视觉字典





基于Bags of features的目标识别

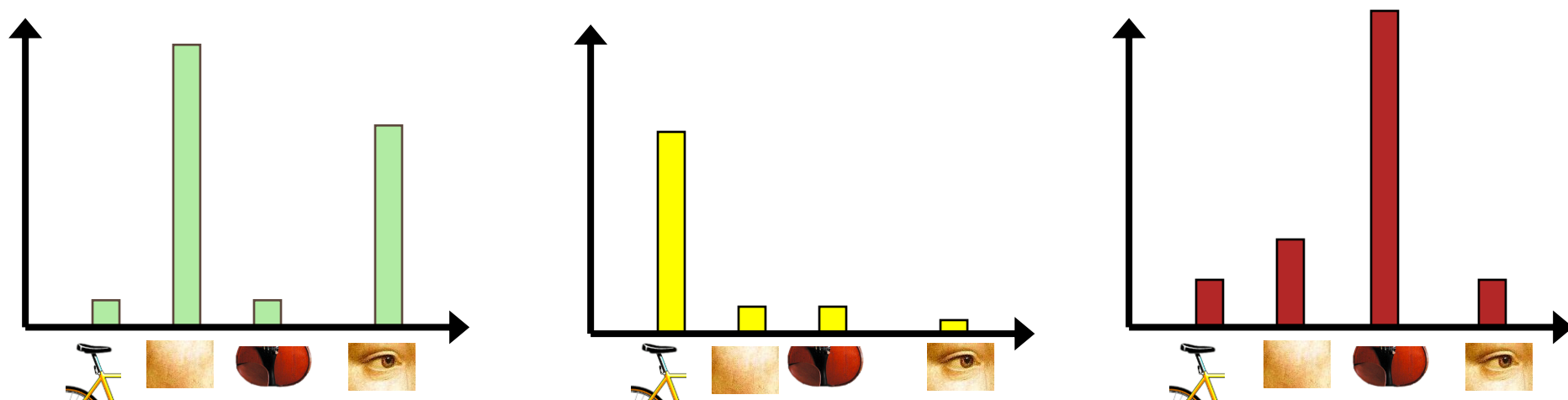
1. 特征提取
2. 学习视觉字典
3. 利用视觉字典对特征进行量化



基于Bags of features的目标识别

1. 特征提取
2. 学习视觉字典
3. 利用视觉字典对特征进行量化
4. 将图像表达为视觉字典出现的频率

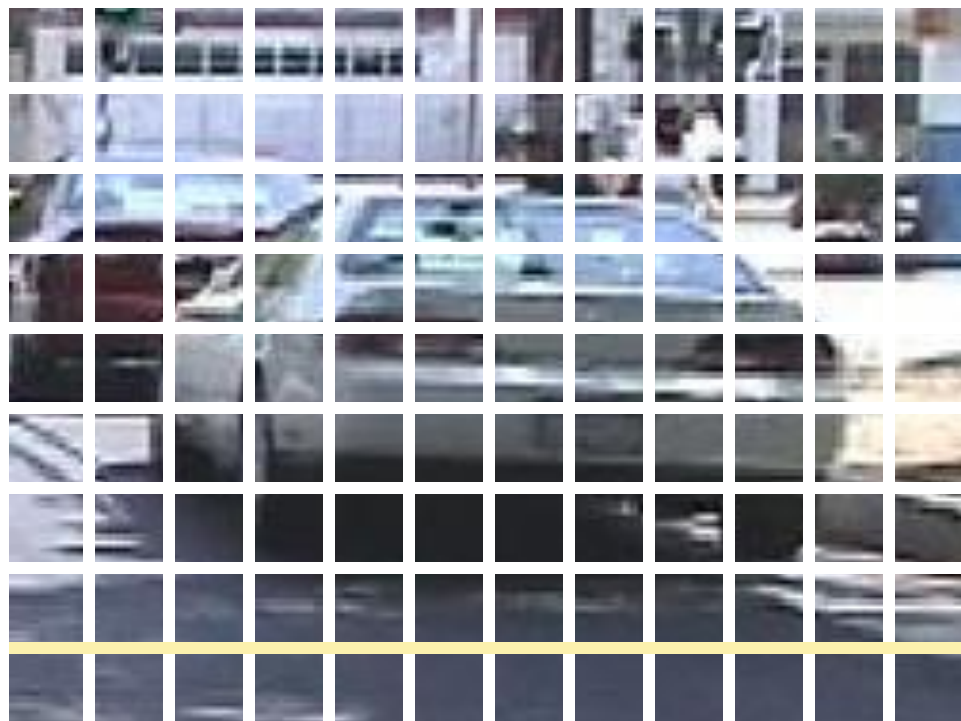
图像表达





特征提取

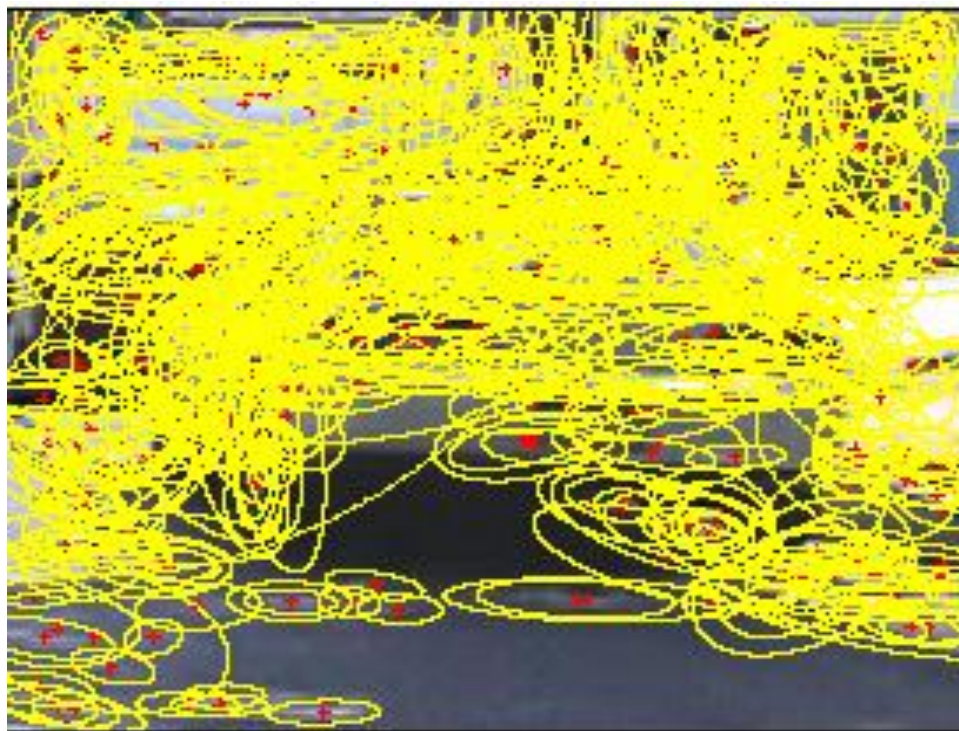
1. 规则的网格区域





特征提取

1. 规则的网格区域
2. 感兴趣点

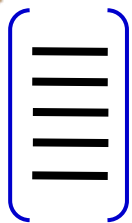




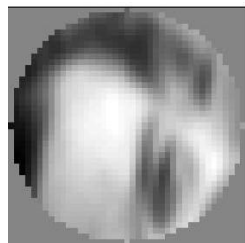
特征提取

1. 规则的网格区域
2. 感兴趣点
3. 其他方法（随机采样、基于分割的区域）

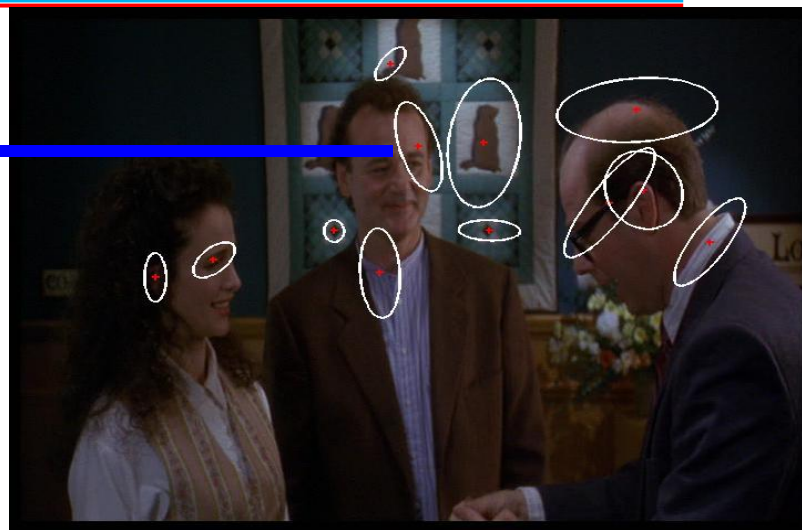
特征提取



计算
SIFT描述子



区域归一化

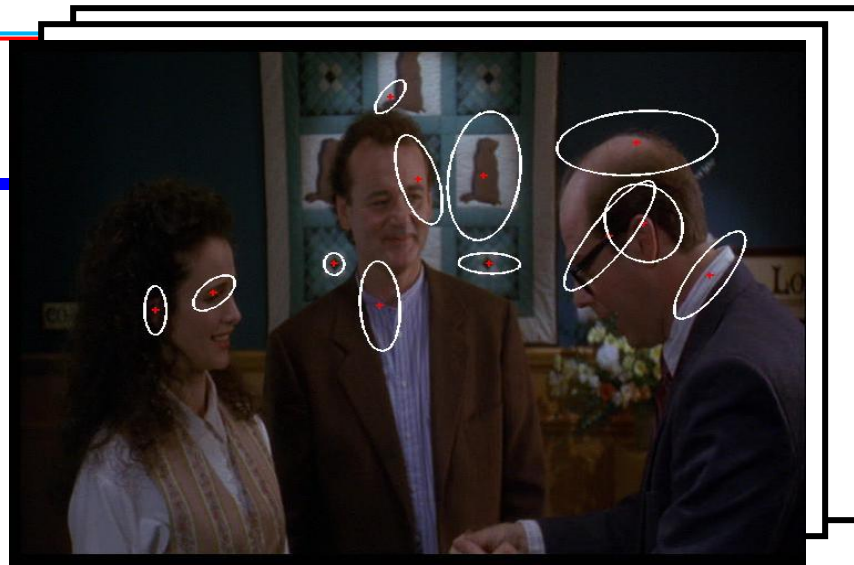


感兴趣点检测



特征提取

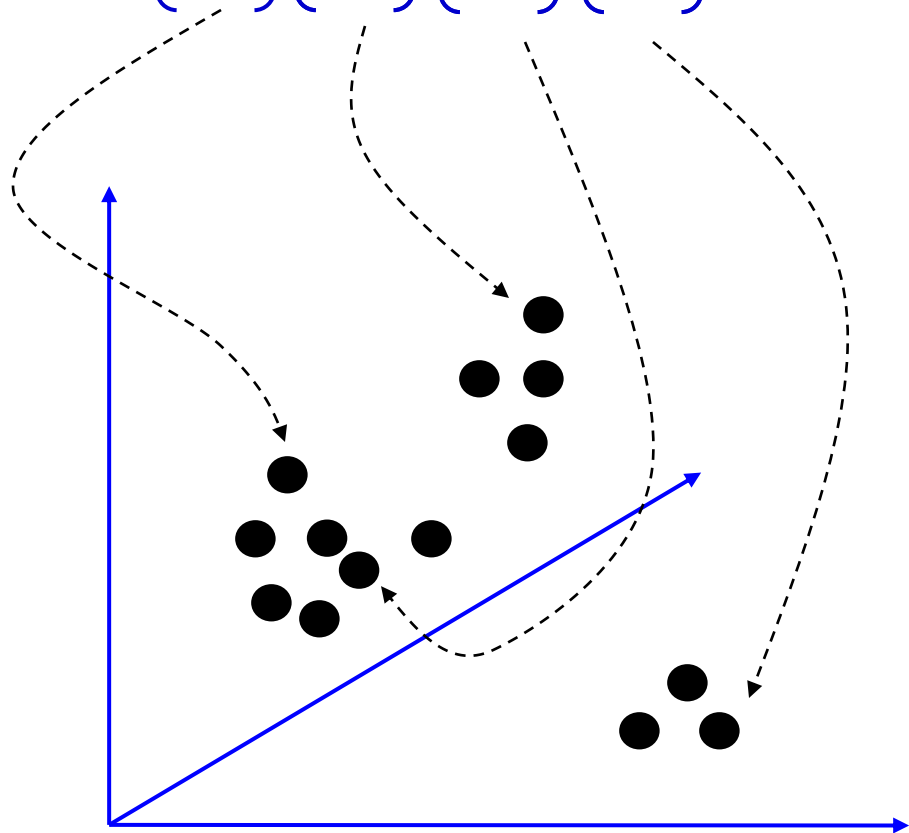
$\left(\begin{array}{c} \text{—} \\ \text{—} \\ \text{—} \end{array} \right) \left(\begin{array}{c} \text{—} \\ \text{—} \\ \text{—} \end{array} \right) \left(\begin{array}{c} \text{—} \\ \text{—} \\ \text{—} \end{array} \right) \left(\begin{array}{c} \text{—} \\ \text{—} \\ \text{—} \end{array} \right) \dots$





学习视觉字典

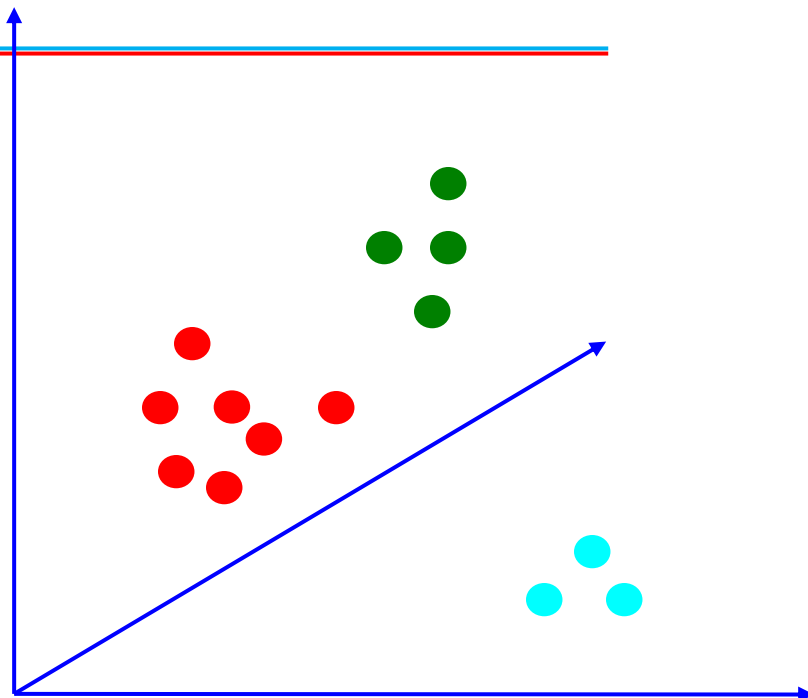
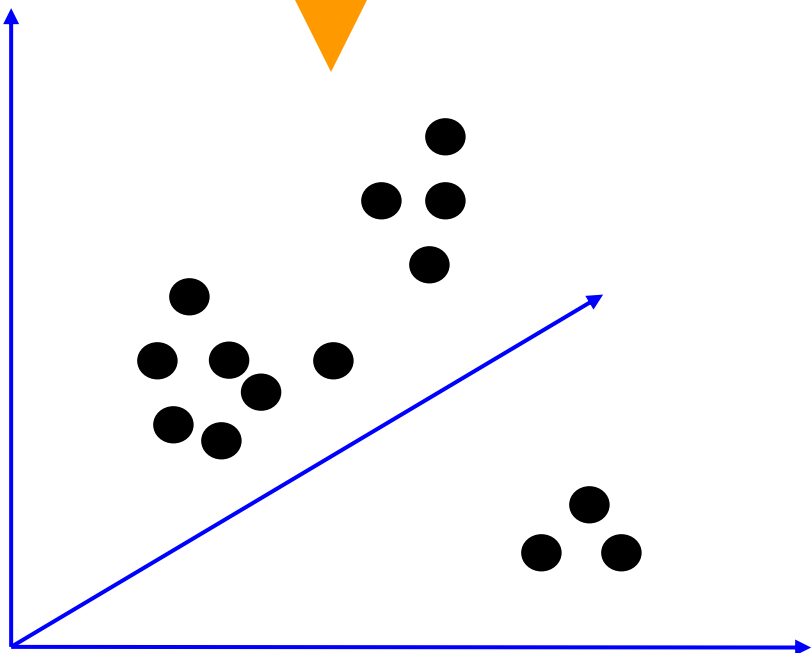
$\left(\begin{smallmatrix} \text{—} \\ \text{—} \\ \text{—} \end{smallmatrix} \right) \left(\begin{smallmatrix} \text{—} \\ \text{—} \\ \text{—} \end{smallmatrix} \right) \left(\begin{smallmatrix} \text{—} \\ \text{—} \\ \text{—} \end{smallmatrix} \right) \left(\begin{smallmatrix} \text{—} \\ \text{—} \\ \text{—} \end{smallmatrix} \right) \dots$





学习视觉字典

(三)(三)(三)(三)...

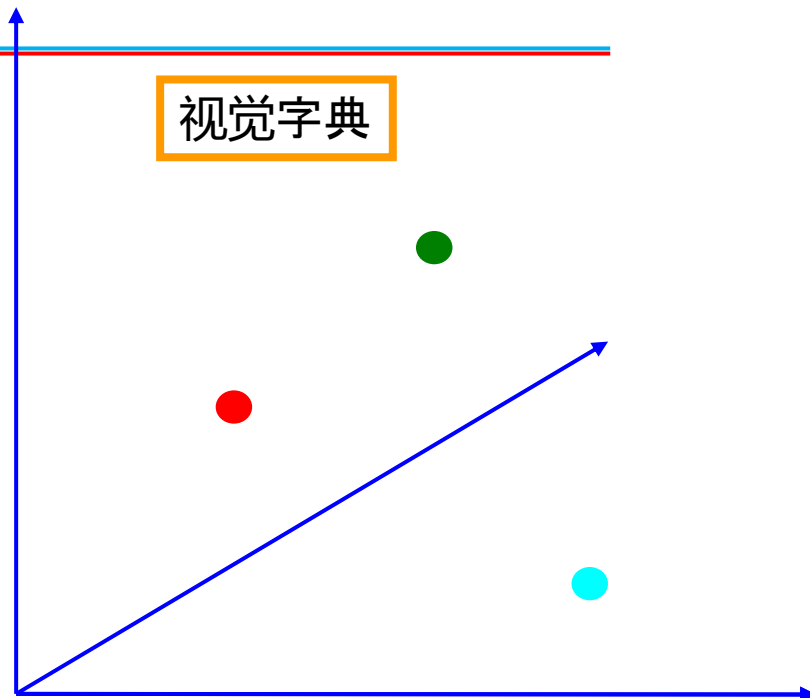
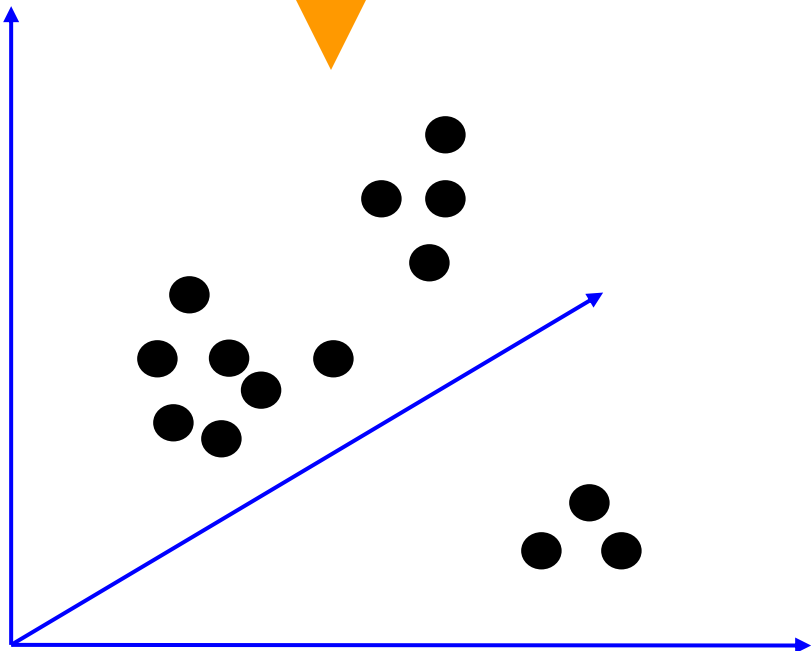


聚类



学习视觉字典

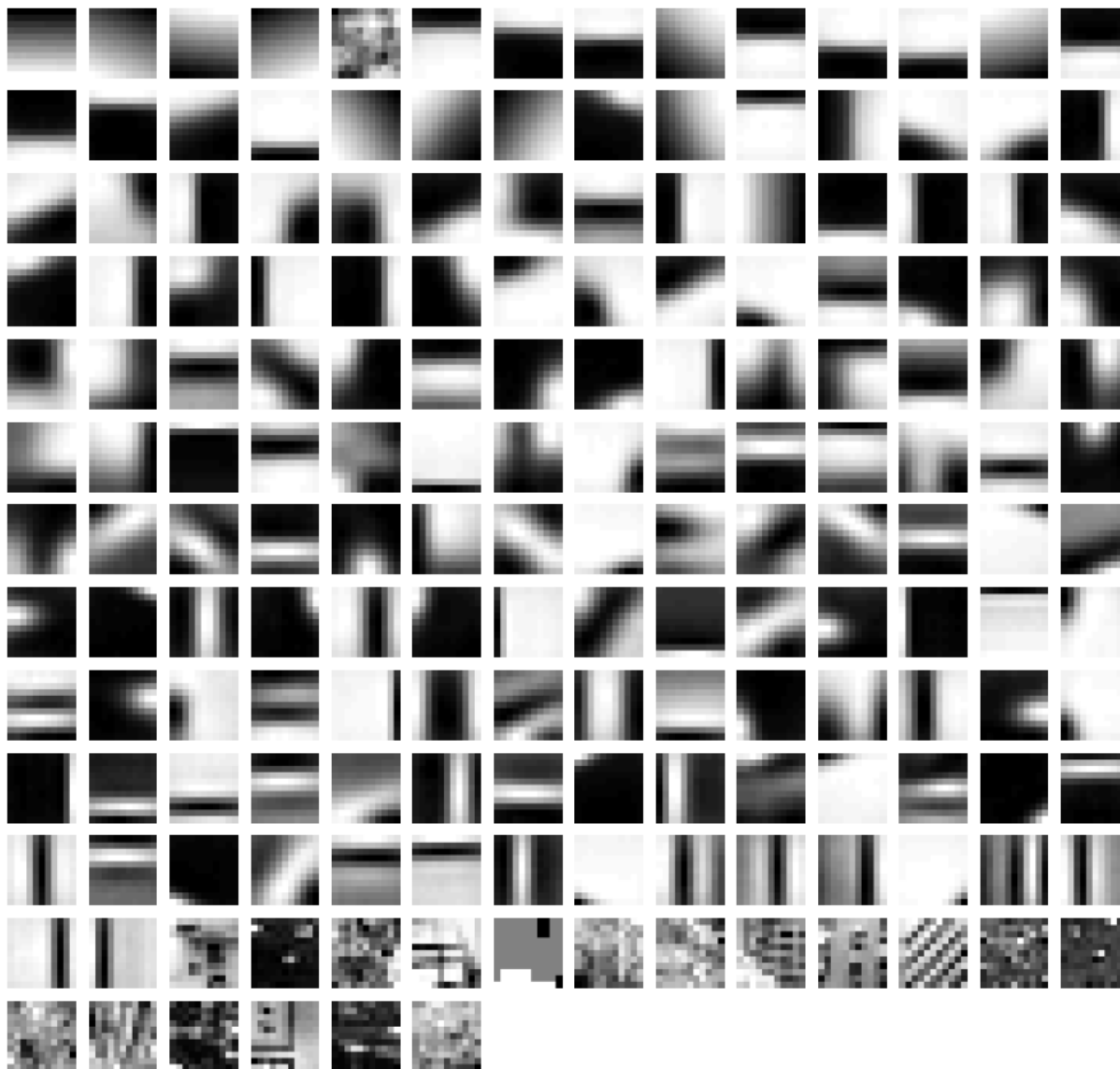
(三)(三)(三)(三)...



聚类

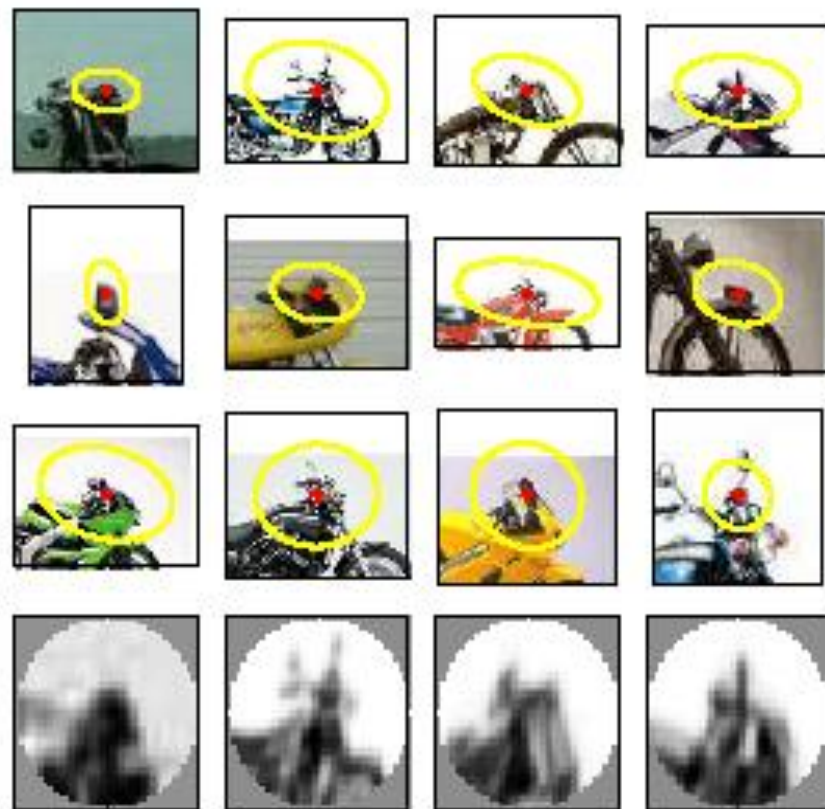
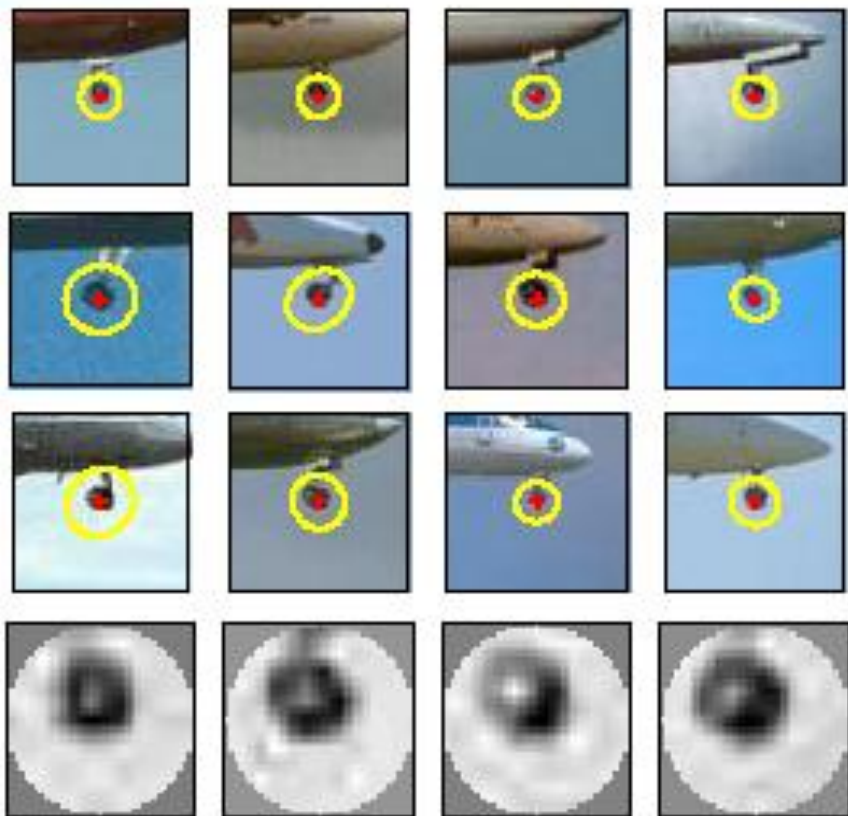


视觉字典举例



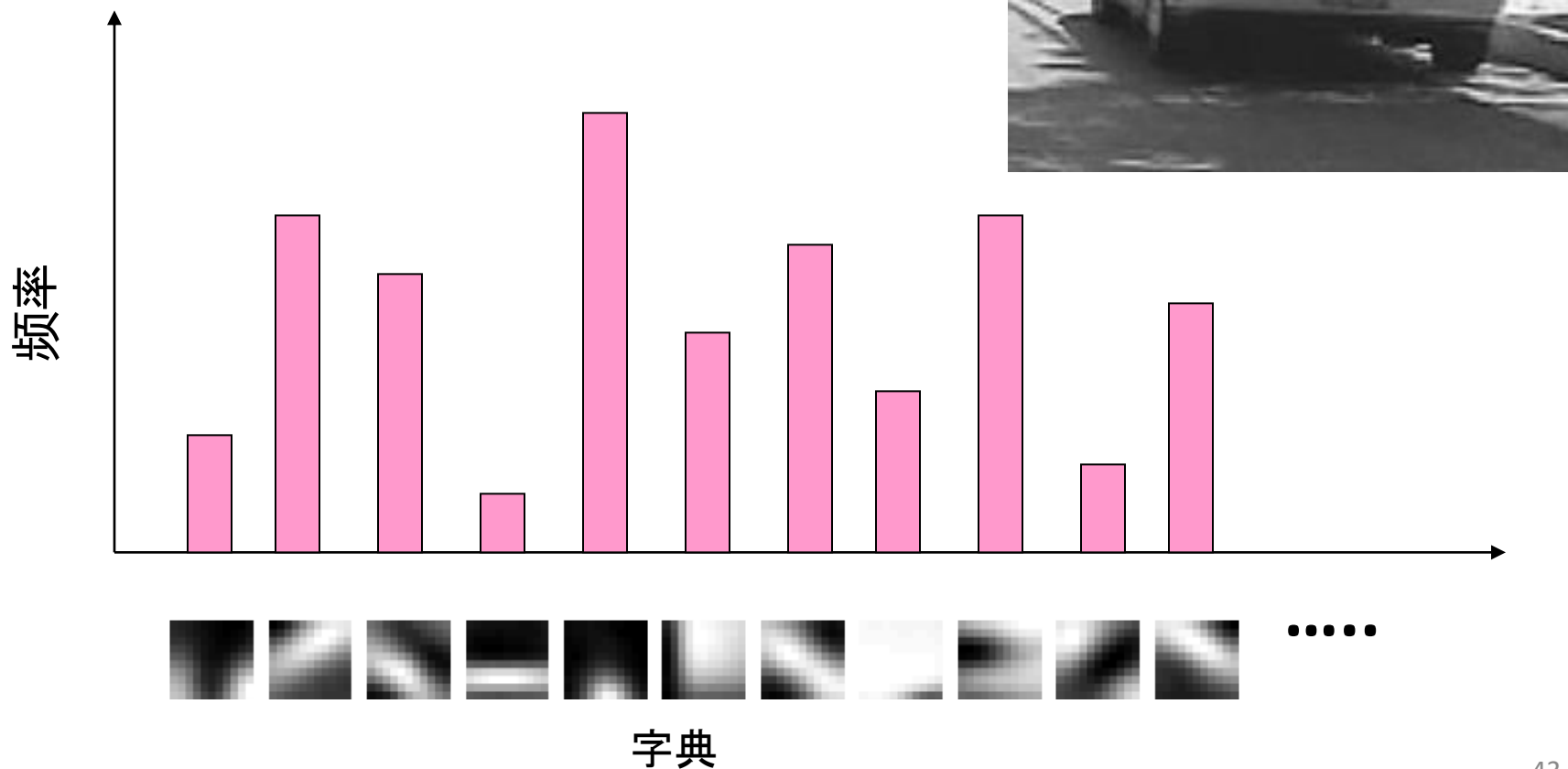


字典对应的图像区域





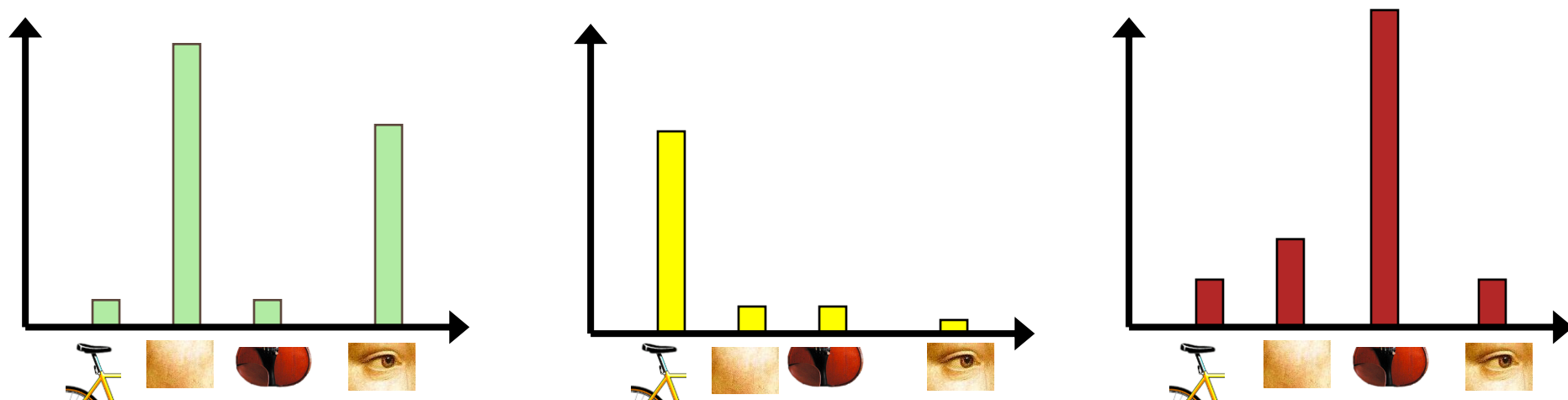
图像表达





图像表达

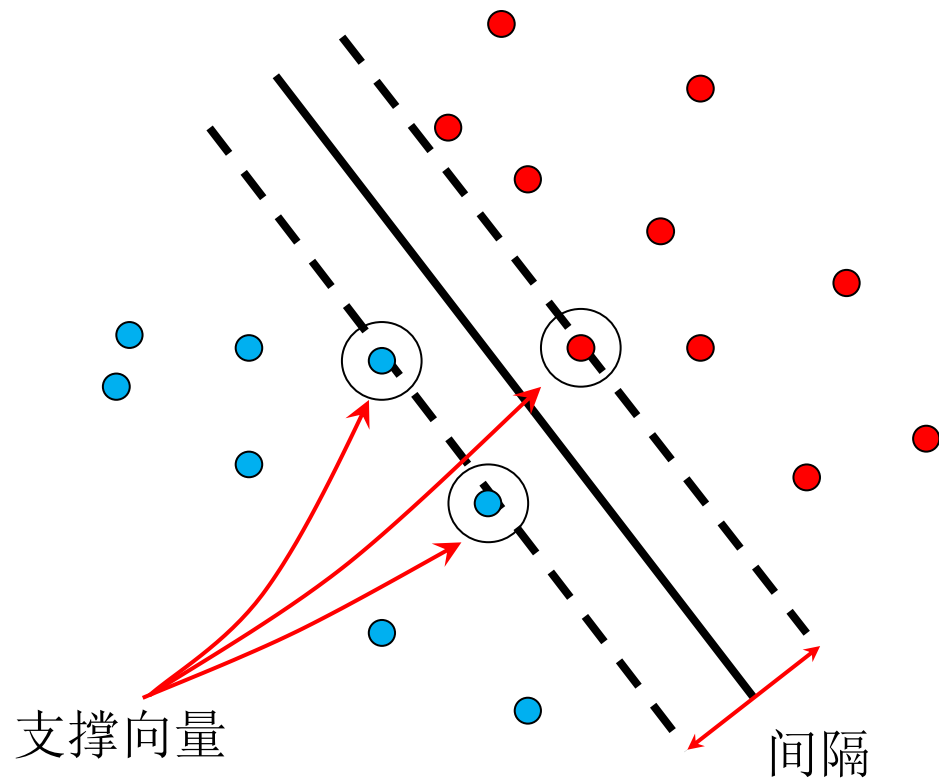
- 得到不同类型目标的bag-of-features的表达后，我们如何对它们进行区分？





目标分类

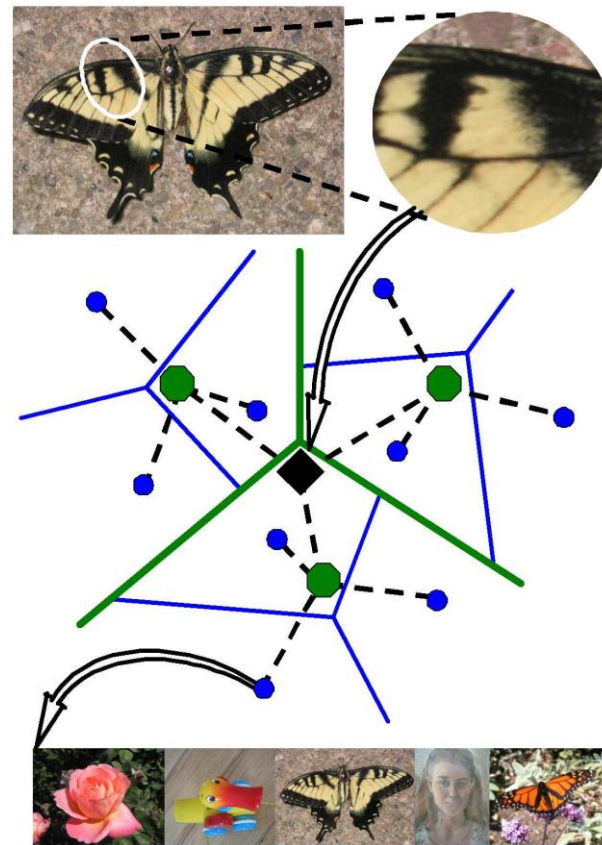
- 产生式与判别式
 - 最近邻分类器
 - K近邻分类器
 - SVM分类器





问题

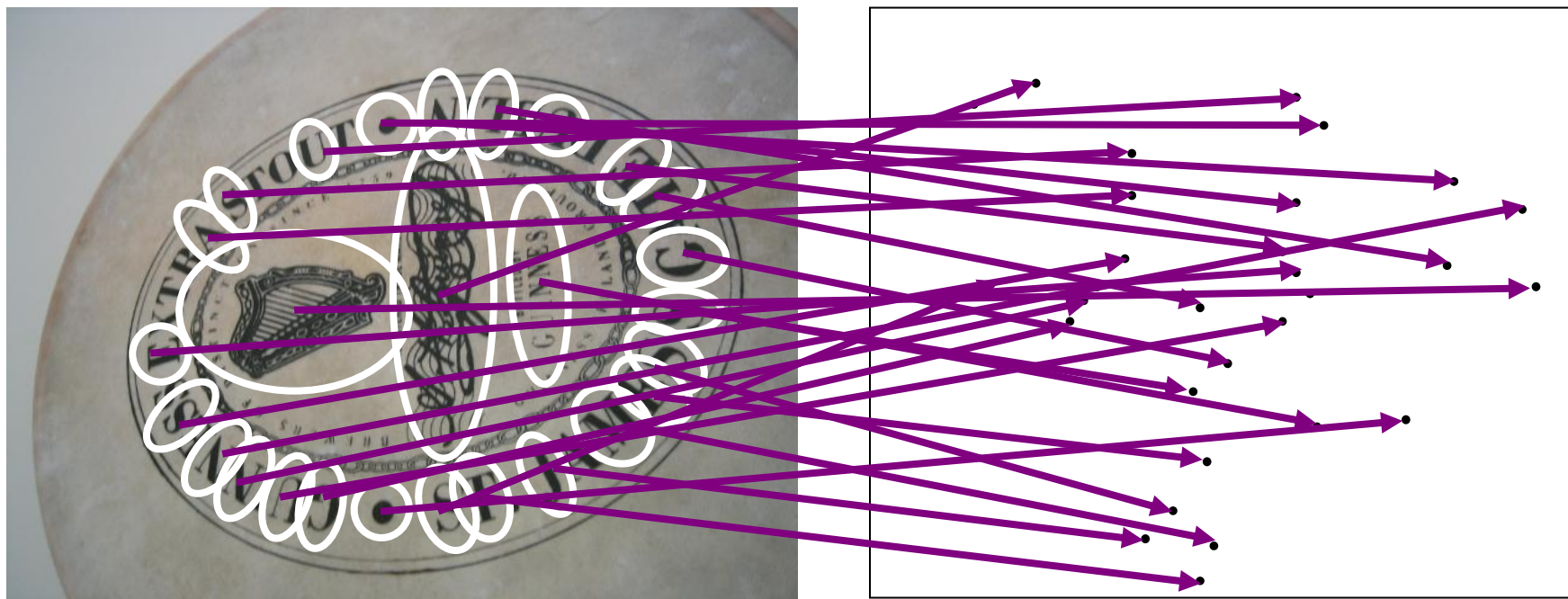
- 如何选择字典大小
- 产生式还是判别式
- 计算效率
 - 字典树
(Scalable Recognition with
a Vocabulary Tree
Nister & Stewenius, 2006)





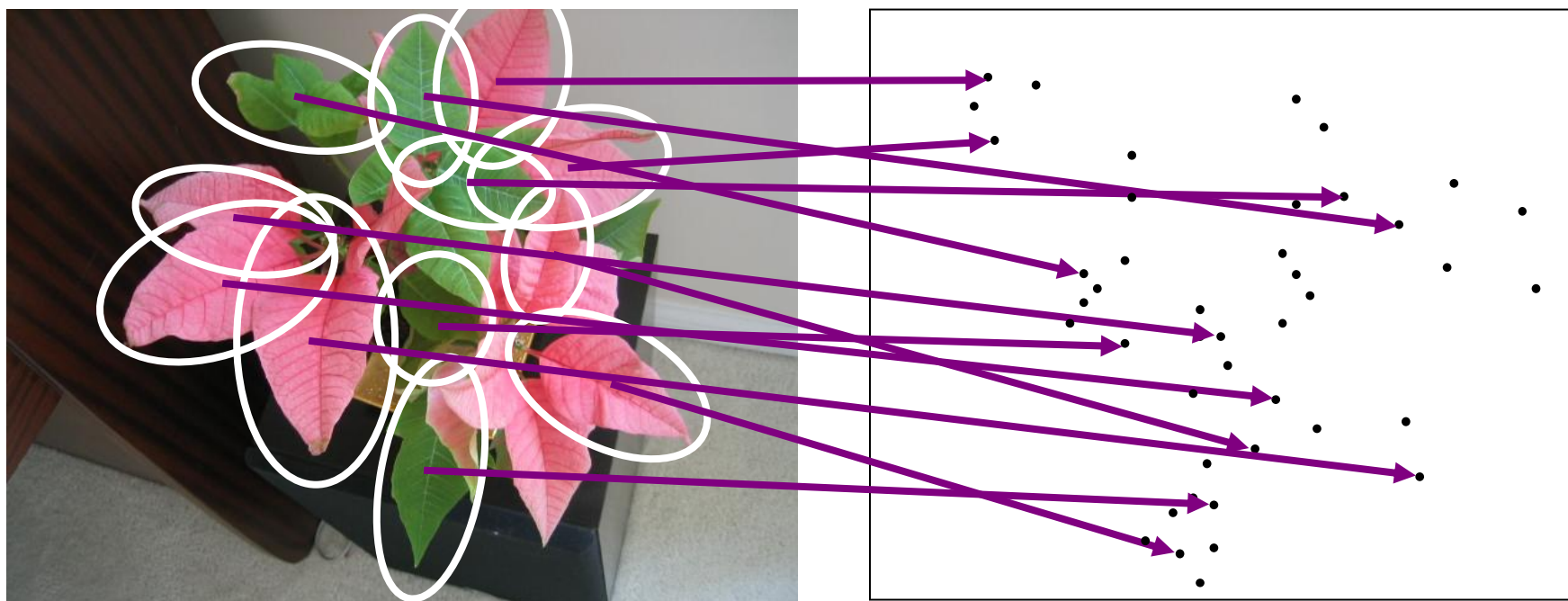
示例：基于字典树的识别

- 提取图像的局部特征



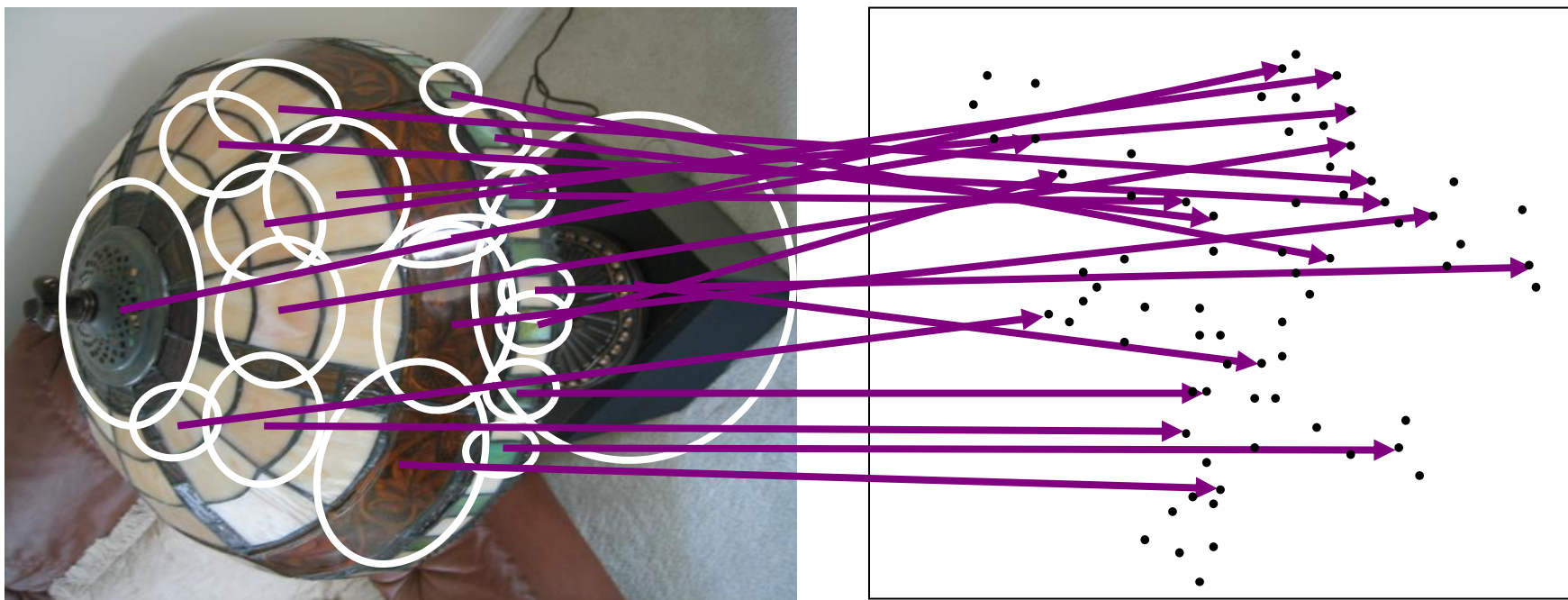


示例：基于字典树的识别



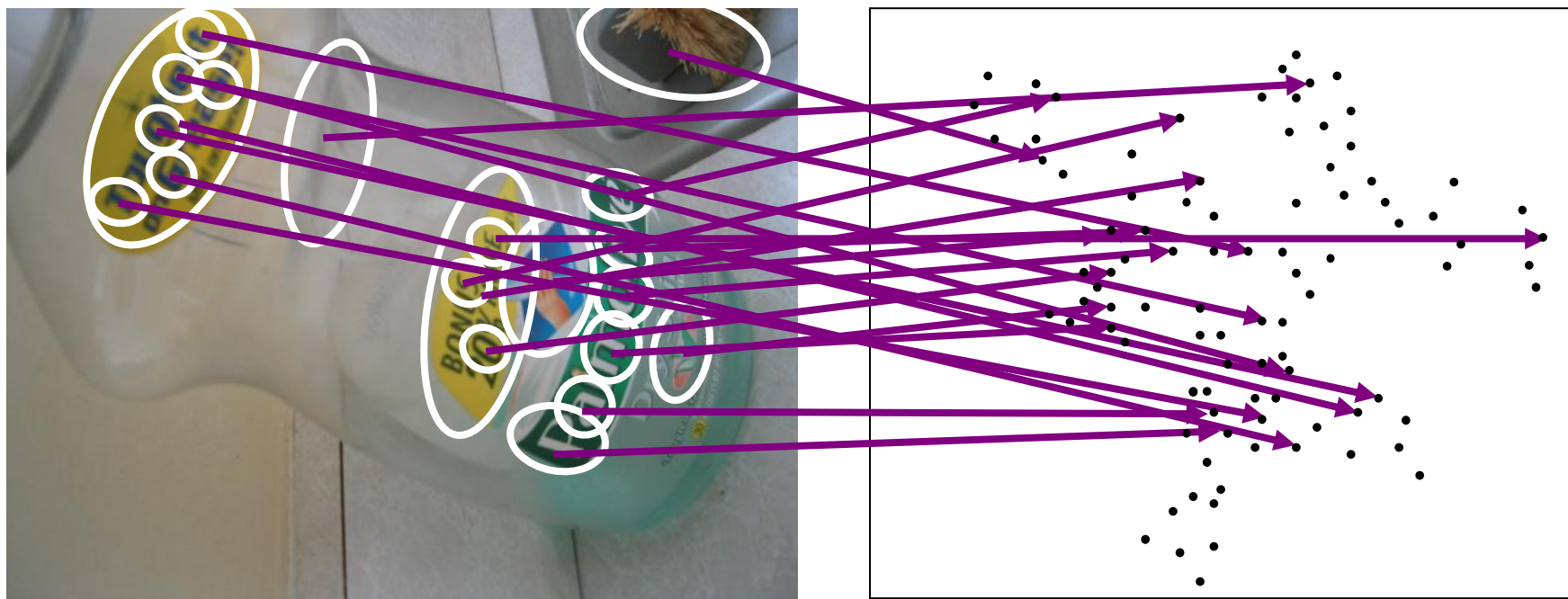


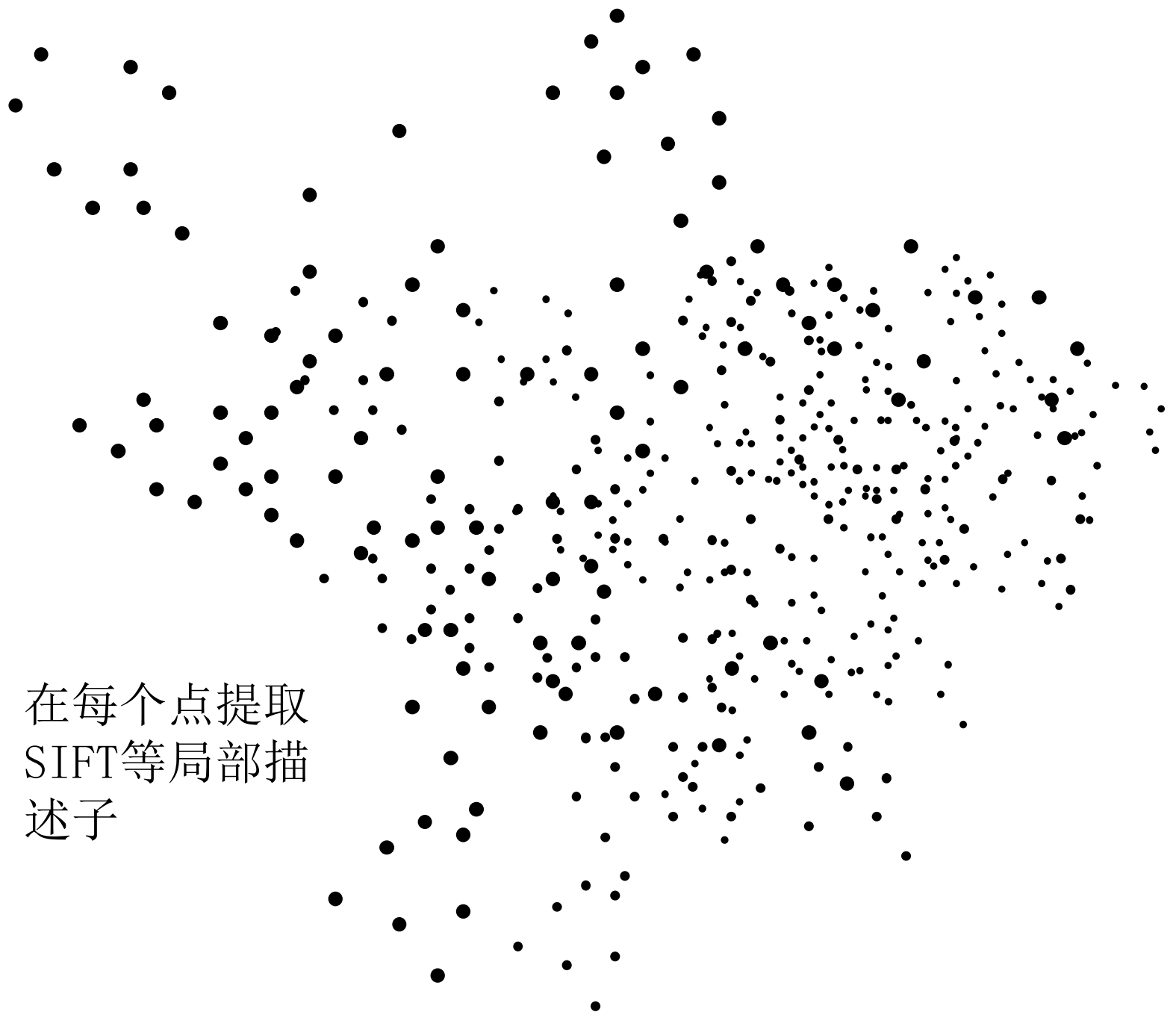
示例：基于字典树的识别



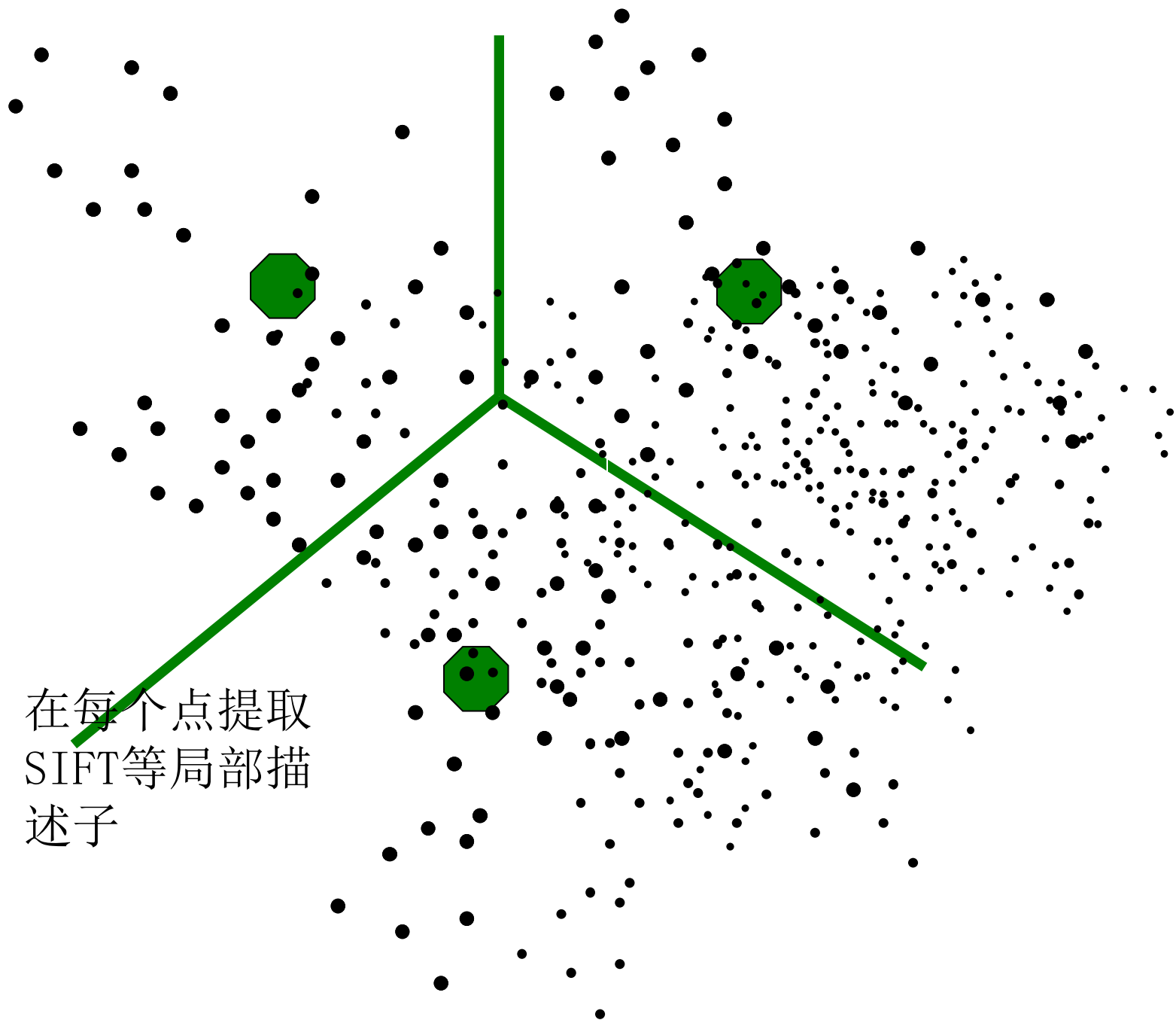


示例：基于字典树的识别





在每个点提取
SIFT等局部描
述子



在每个点提取
SIFT等局部描
述子



示例：基于字典树的识别

- 视觉单词
- 每一组属于一个语义

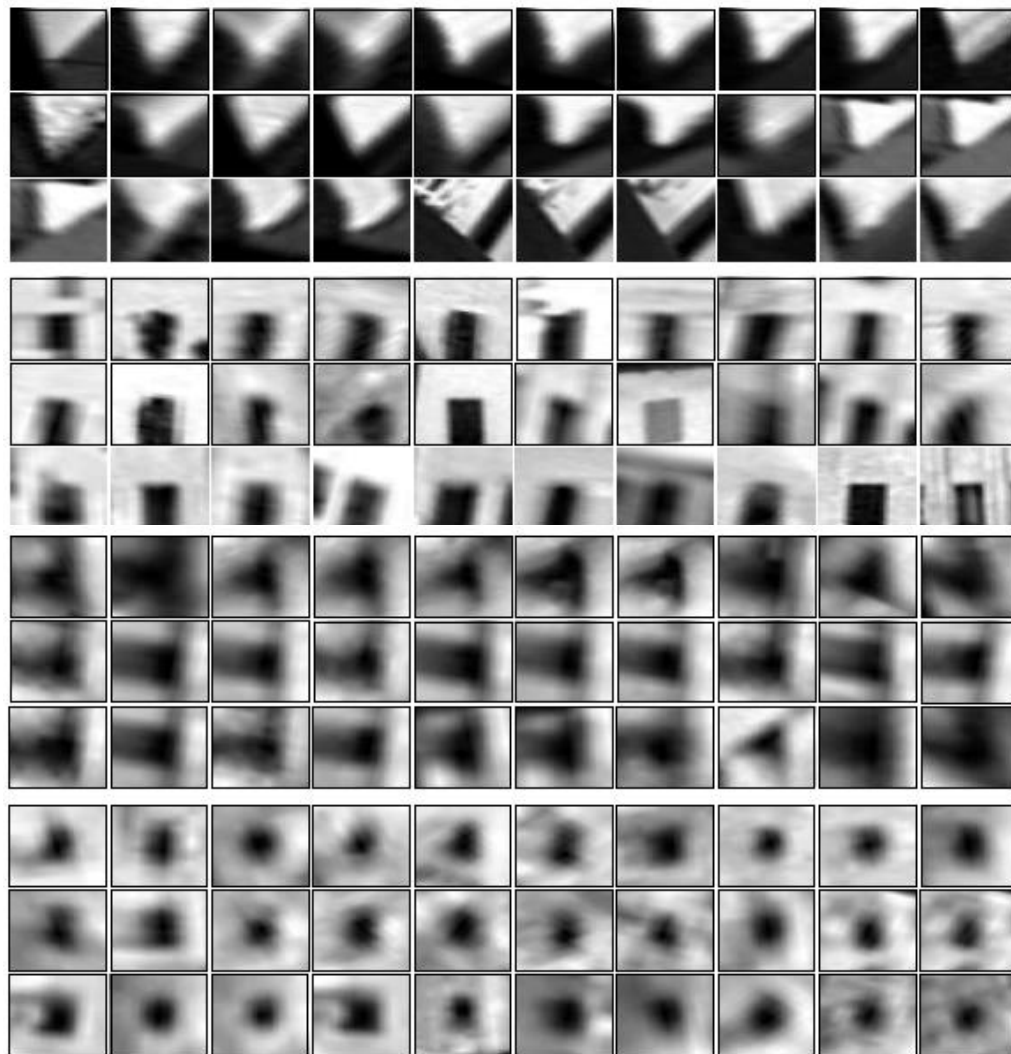
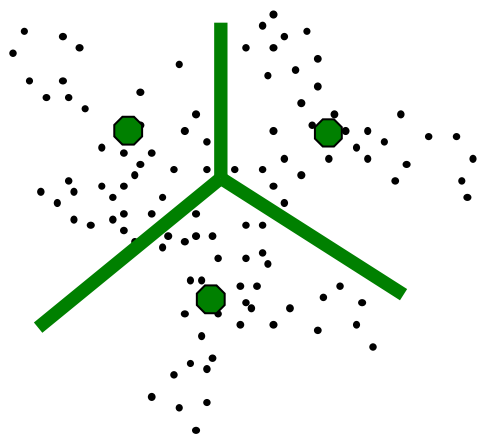
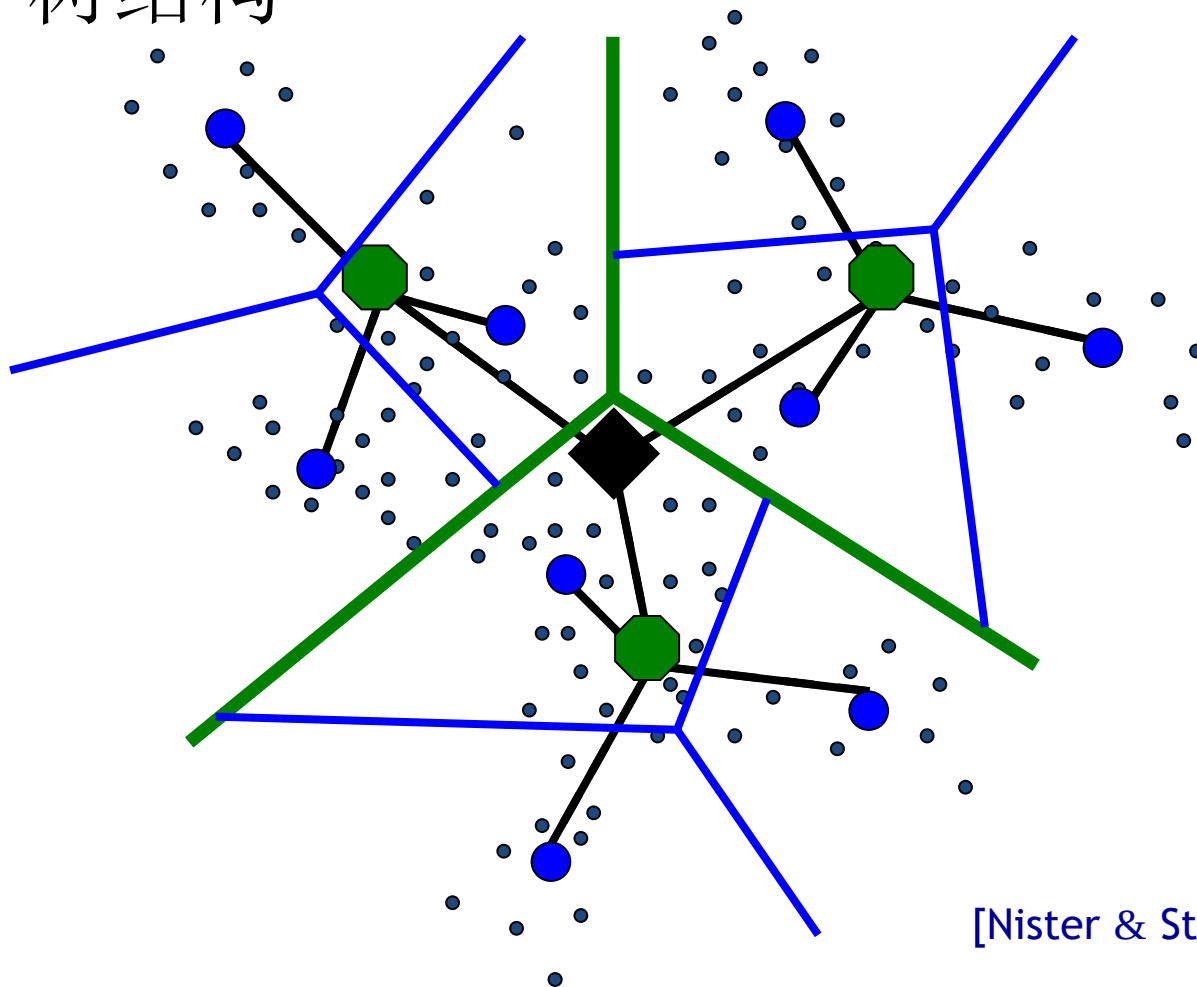


Figure from Sivic & Zisserman, ICCV 2003



示例：基于字典树的识别

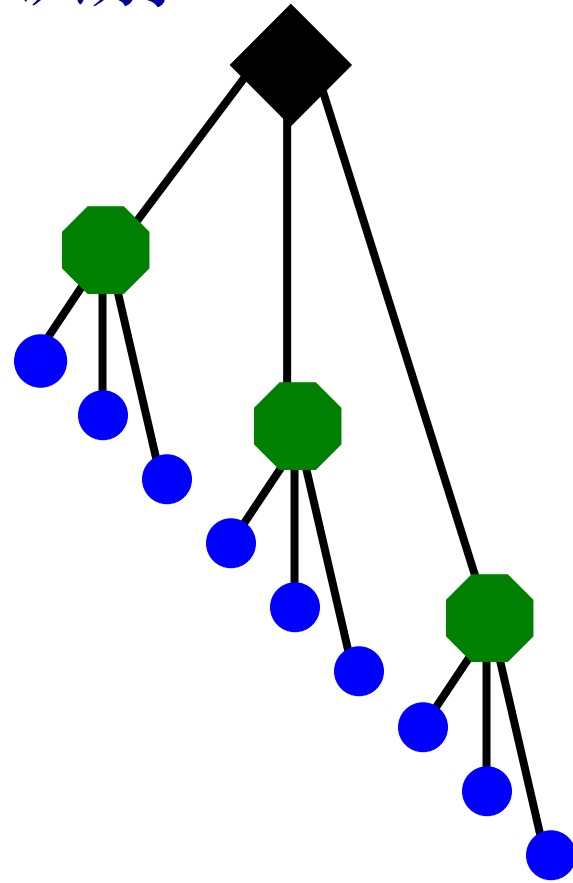
- 树结构



[Nister & Stewenius, CVPR'06]

示例：基于字典树的识别

- 字典树
- 训练：完成树结构

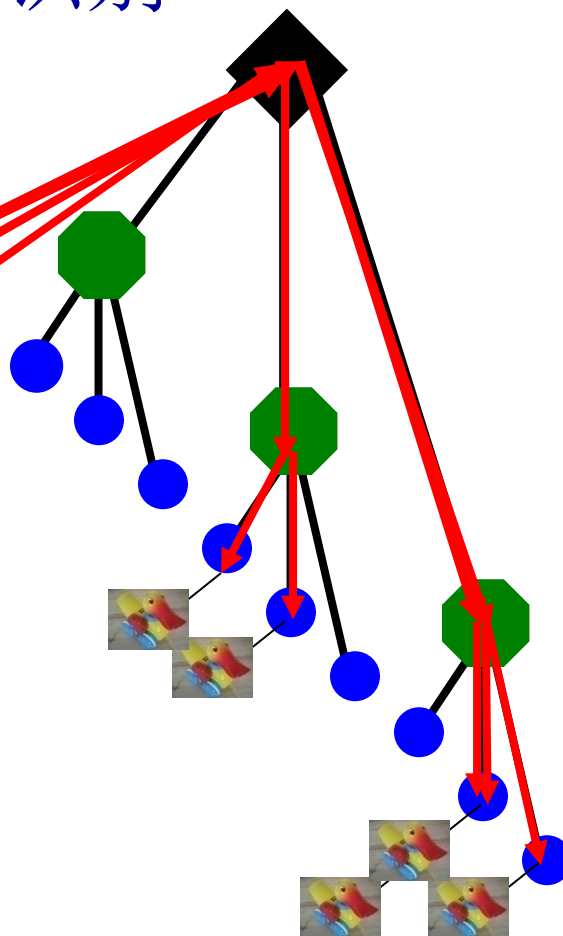


[Nister & Stewenius, CVPR'06]

示例：基于字典树的识别

- 字典树

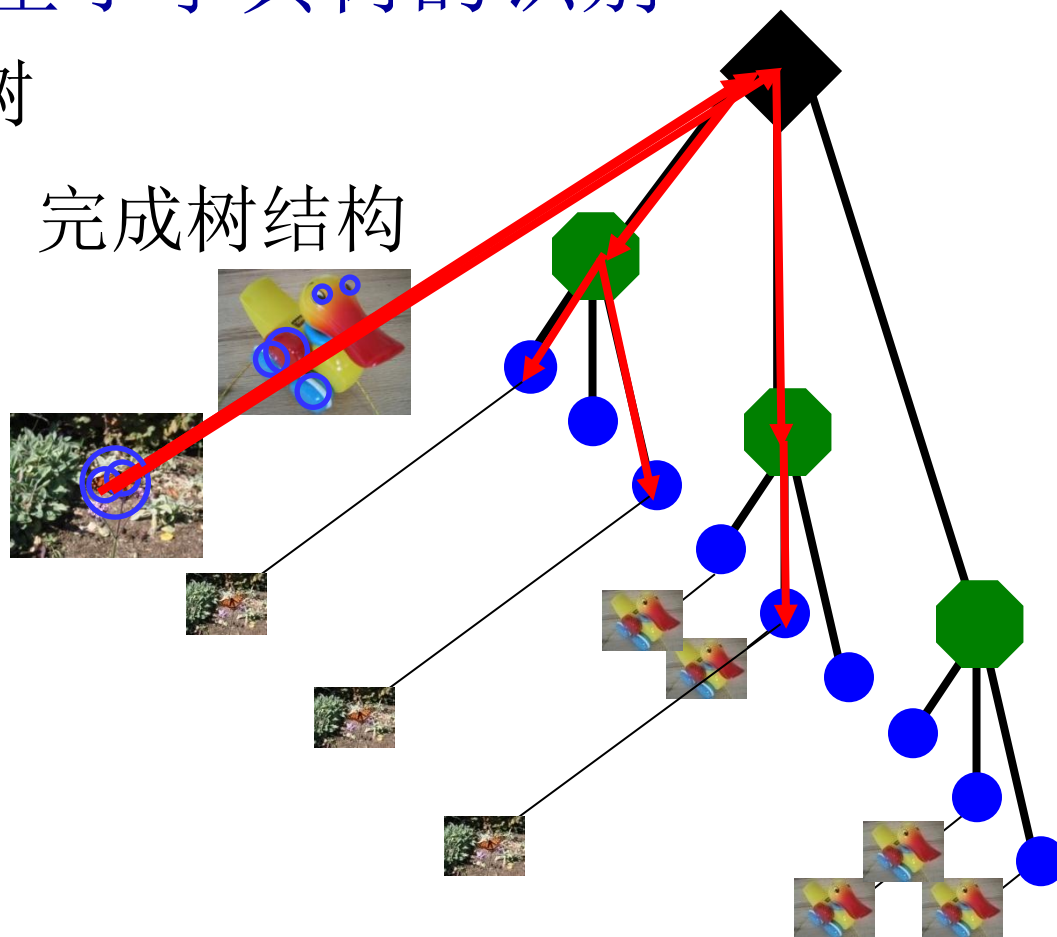
- 训练：完成树结构



[Nister & Stewenius, CVPR'06]

示例：基于字典树的识别

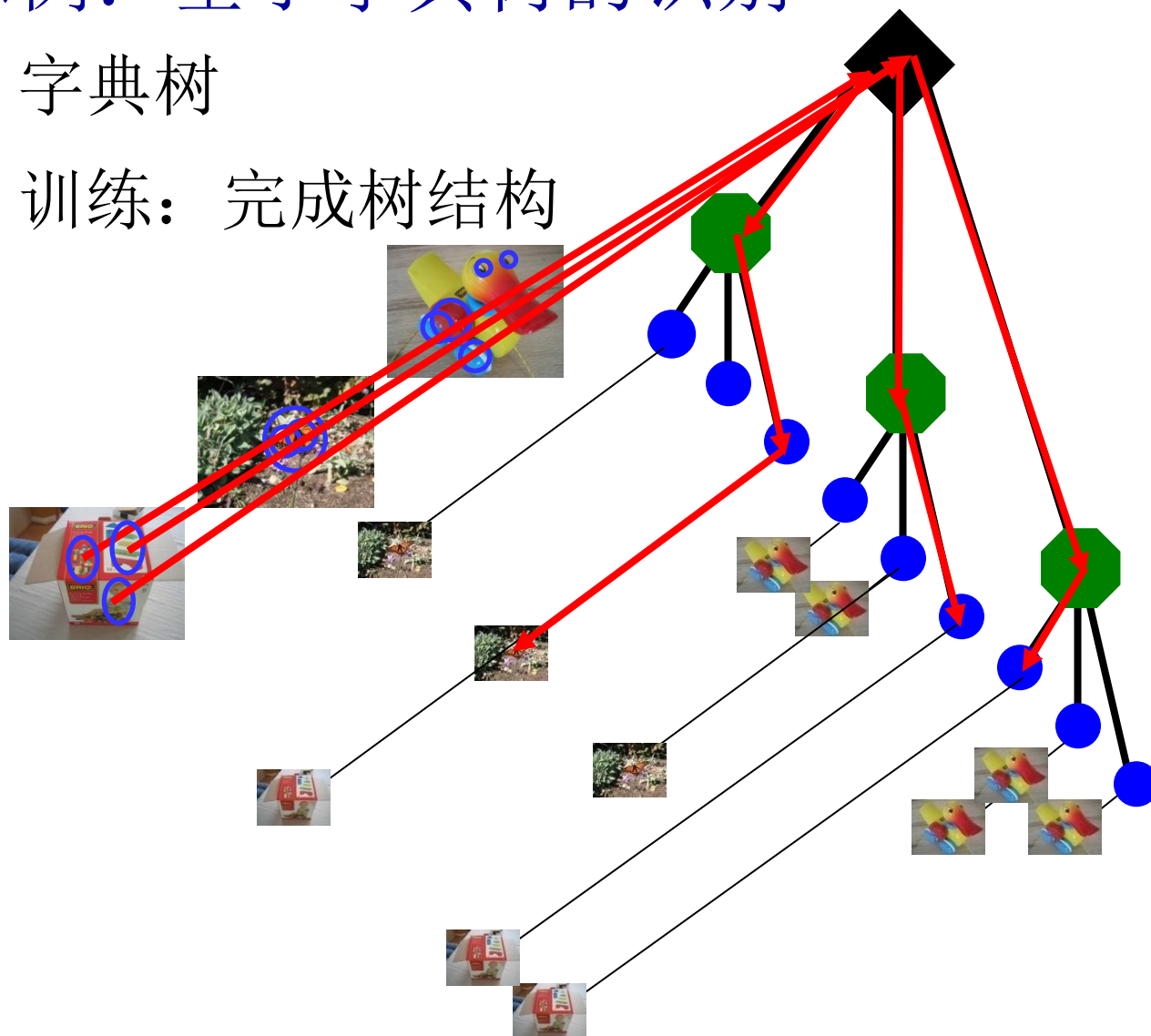
- 字典树
- 训练：完成树结构



[Nister & Stewenius, CVPR'06]

示例：基于字典树的识别

- 字典树
- 训练：完成树结构

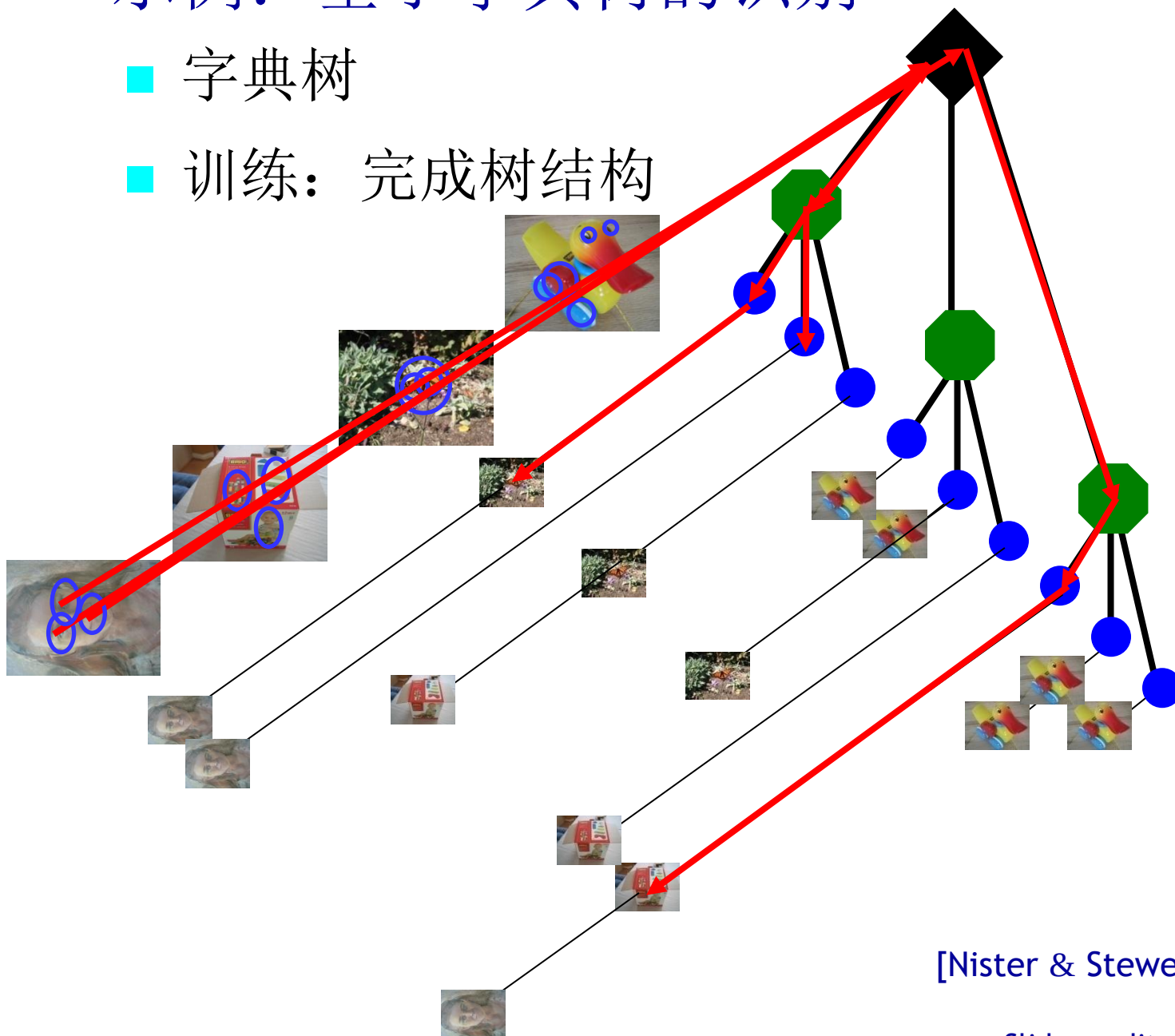


[Nister & Stewenius, CVPR'06]

示例：基于字典树的识别

■ 字典树

■ 训练：完成树结构



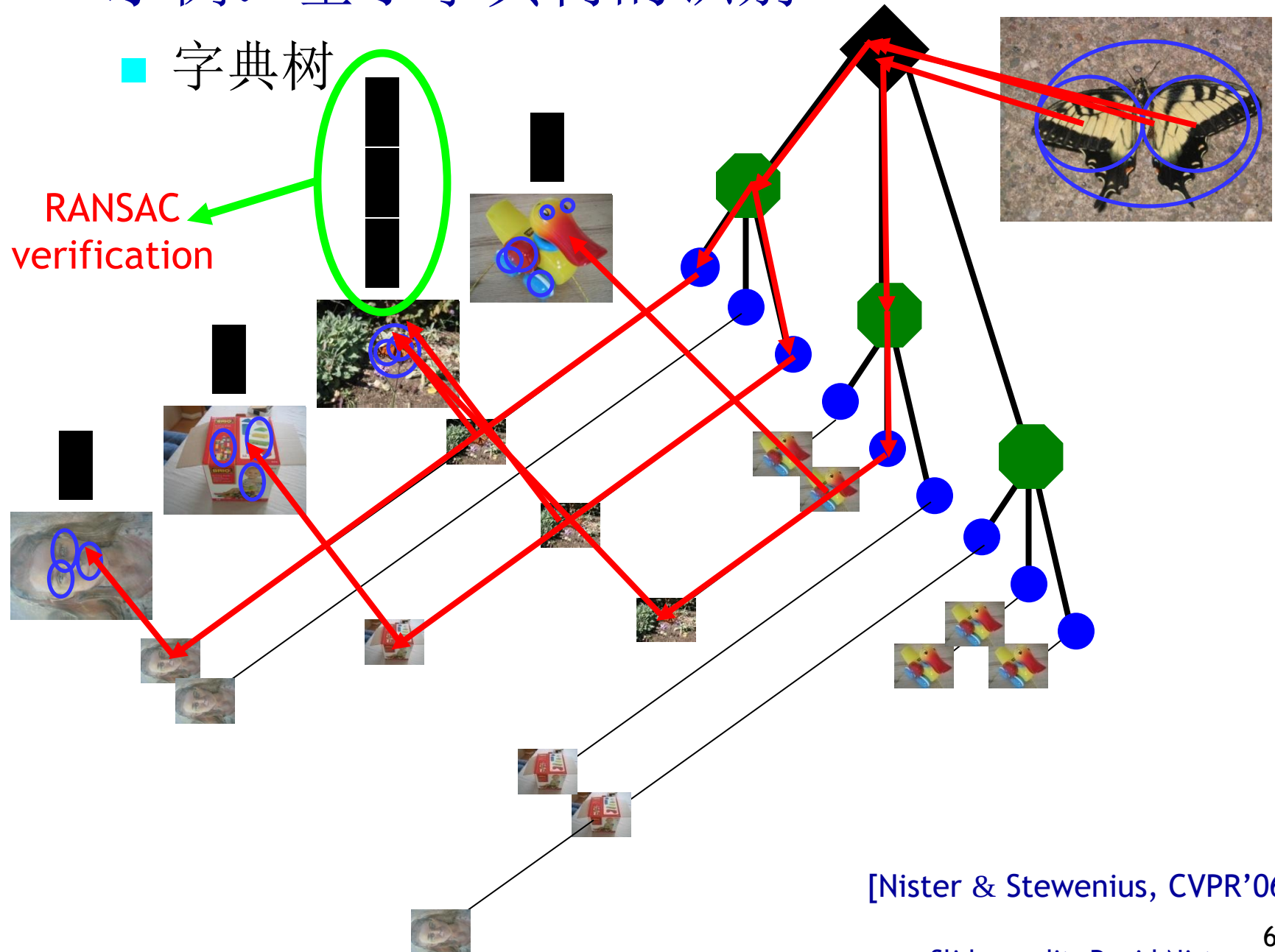
[Nister & Stewenius, CVPR'06]

Slide credit: David Nister

示例：基于字典树的识别

字典树

RANSAC
verification



[Nister & Stewenius, CVPR'06]

Slide credit: David Nister

示例：基于字典树的识别

■ 节点权重设置问题

$$w_i = \ln \frac{N}{N_i},$$

■ 距离度量问题

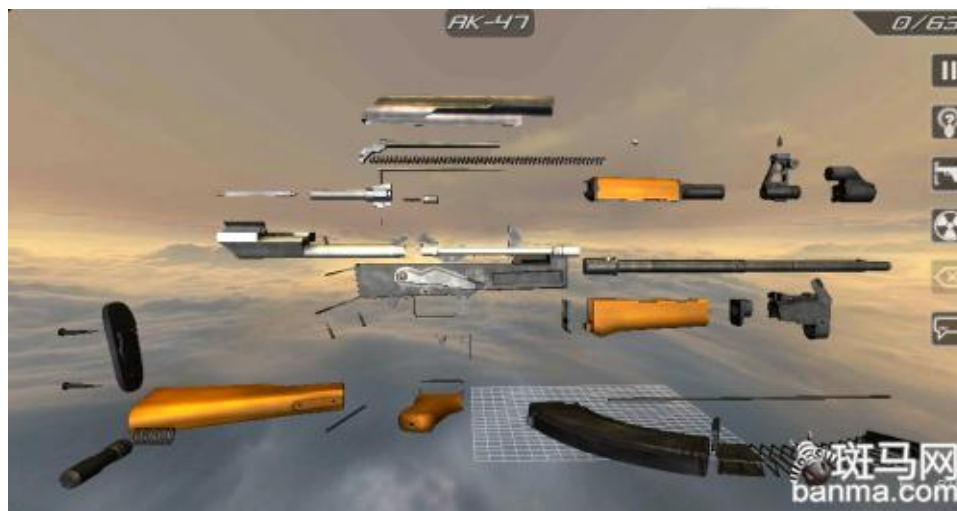
$$\begin{aligned}\|\mathbf{q}-\mathbf{d}\|_p^p &= \sum_i |q_i - d_i|^p \\ &= \sum_{i|d_i=0} |q_i|^p + \sum_{i|q_i=0} |d_i|^p + \sum_{i|q_i \neq 0, d_i \neq 0} |q_i - d_i|^p \\ &= \|\mathbf{q}\|_p^p + \|\mathbf{d}\|_p^p + \sum_{i|q_i \neq 0, d_i \neq 0} (|q_i - d_i|^p - |q_i|^p - |d_i|^p) \\ &= 2 + \sum_{i|q_i \neq 0, d_i \neq 0} (|q_i - d_i|^p - |q_i|^p - |d_i|^p),\end{aligned}$$

$$\|\mathbf{q}-\mathbf{d}\|_2^2 = 2 - 2 \sum_{i|q_i \neq 0, d_i \neq 0} q_i d_i,$$



基于特征点的目标检测方法示例

- Bag-of-words框架
- 缺点？





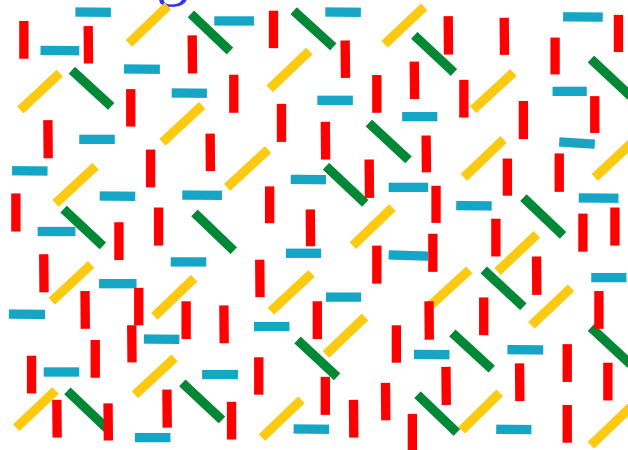
基于特征点的目标检测方法示例

- Bag-of-words框架
- 缺点？
- 如何引入空间信息
- 特征点之间的空间关系
 - 星群模型
 - 空间金字塔模型



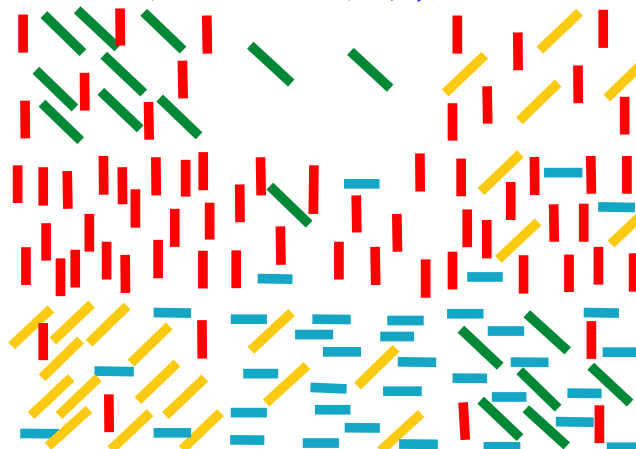
基于特征点的目标检测方法示例

Bag of words 模型

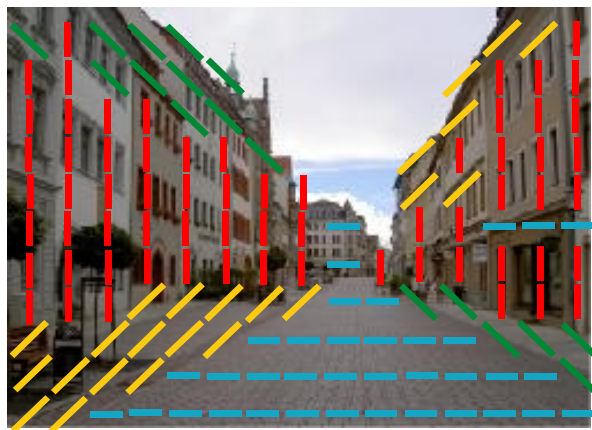


65 17 23 36

加入空间信息

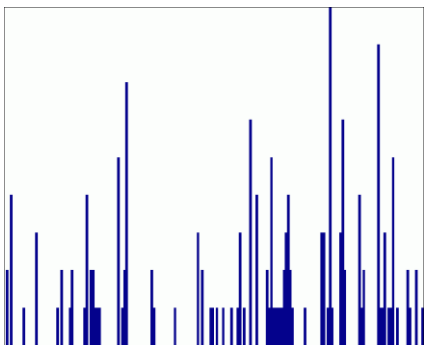


7	8	0	0	0	2	0	0	7	0	4	0
20	0	0	0	11	1	0	2	14	0	3	3
3	0	12	4	0	0	4	16	3	6	0	11



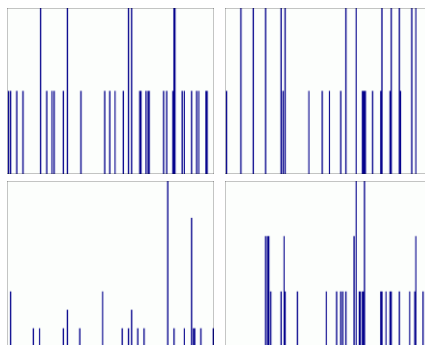
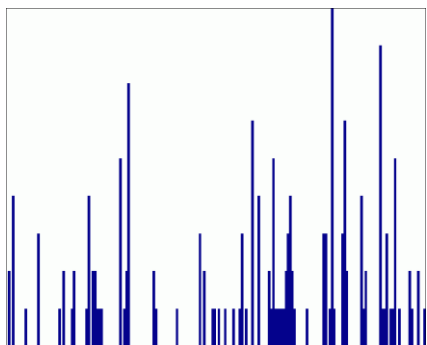
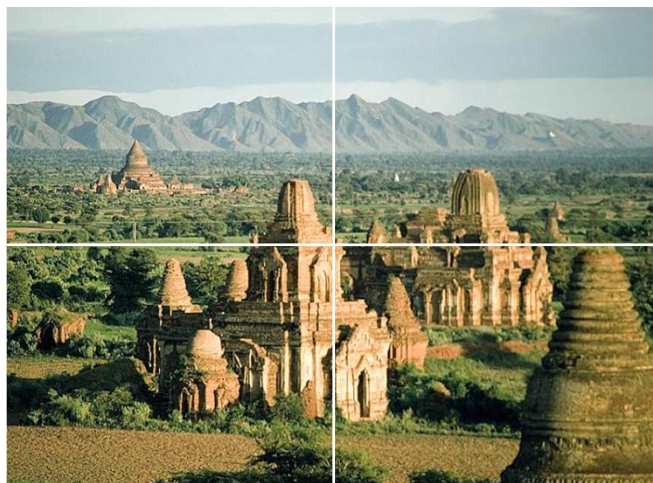


空间金字塔表达



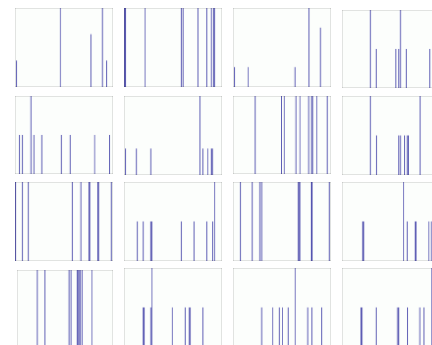
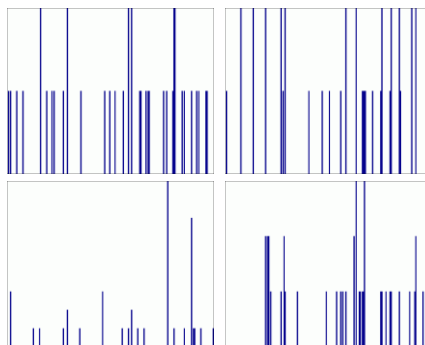
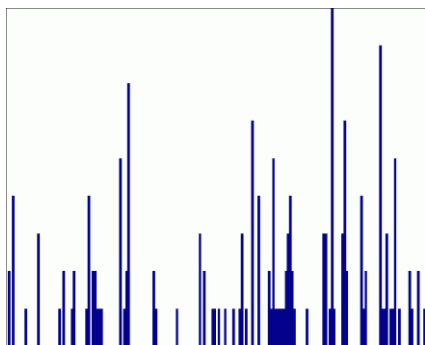
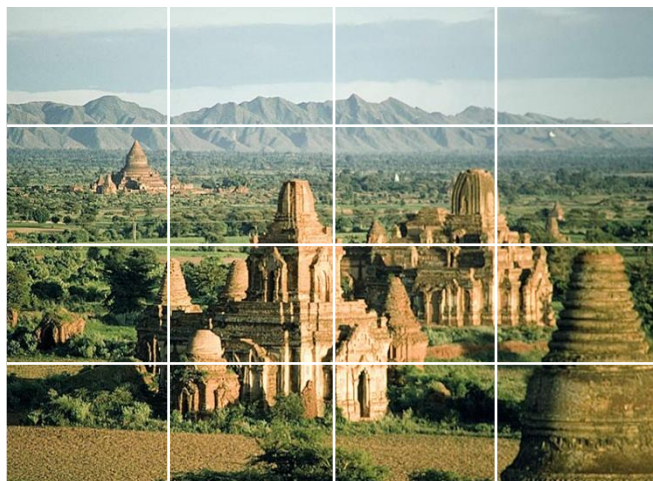


空间金字塔表达





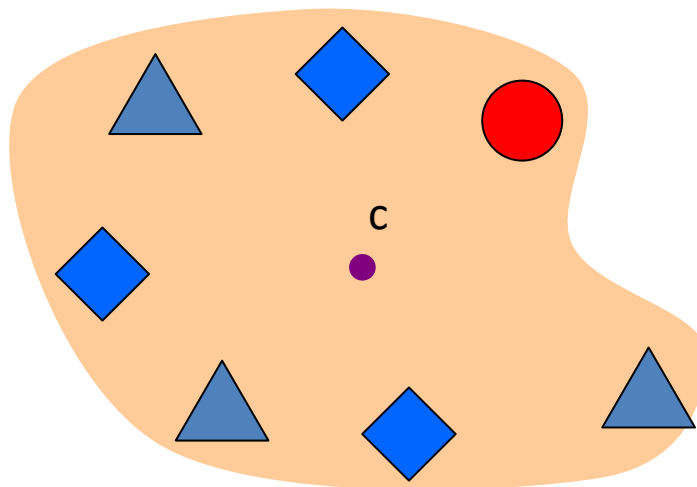
空间金字塔表达





通用霍夫变换——模型构建

Template



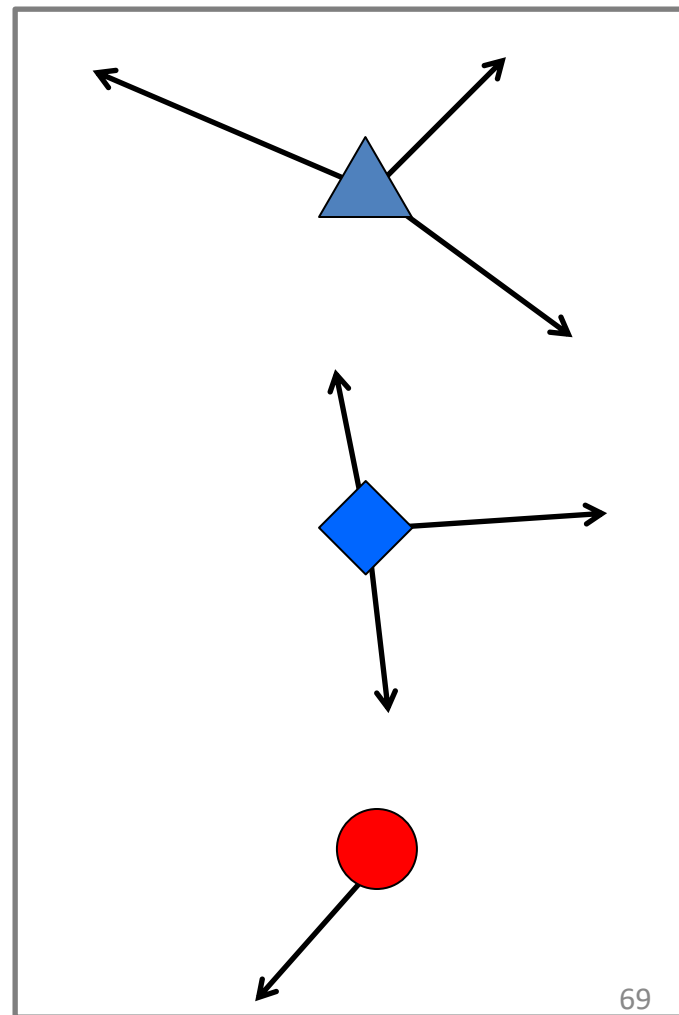
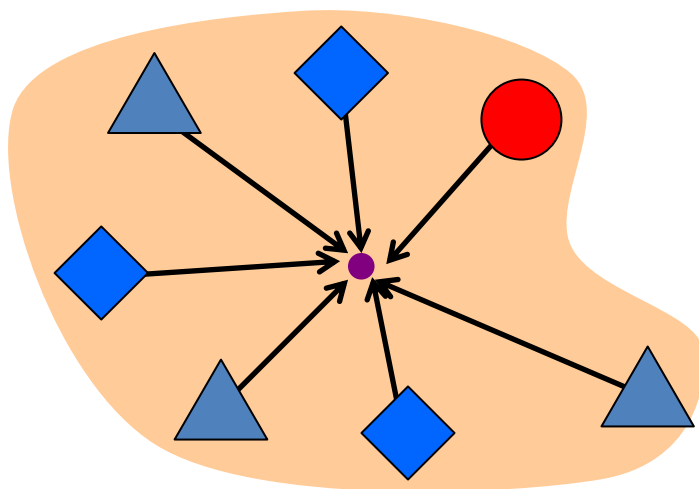


通用霍夫变换——模型构建

Model

对一个类别，用特征，特征相对目标中心的偏移矢量建模

Template

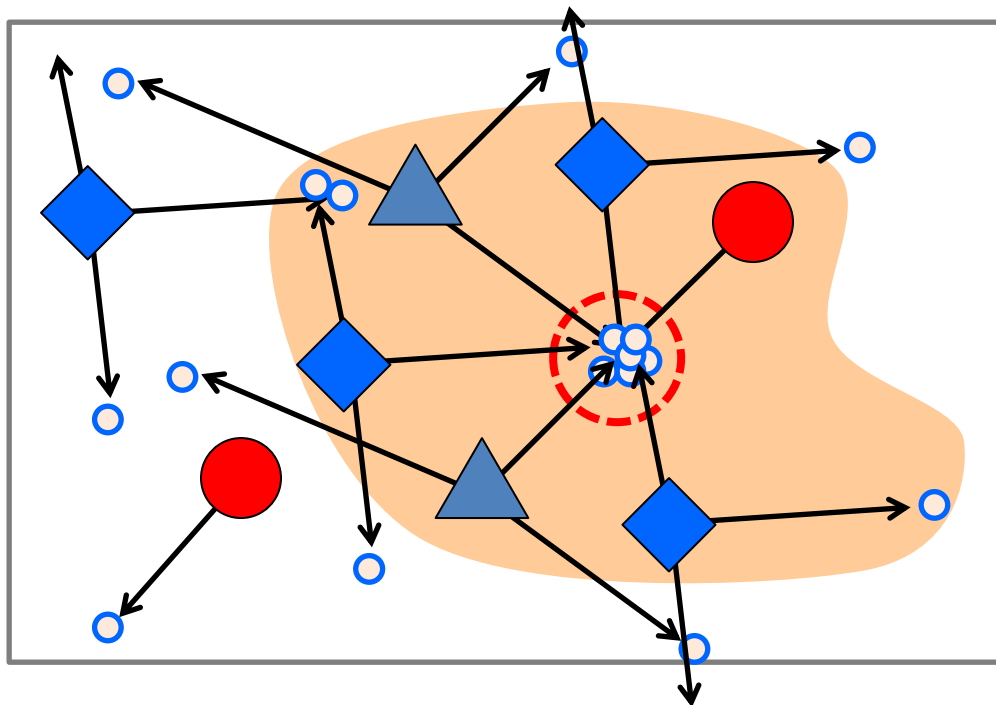




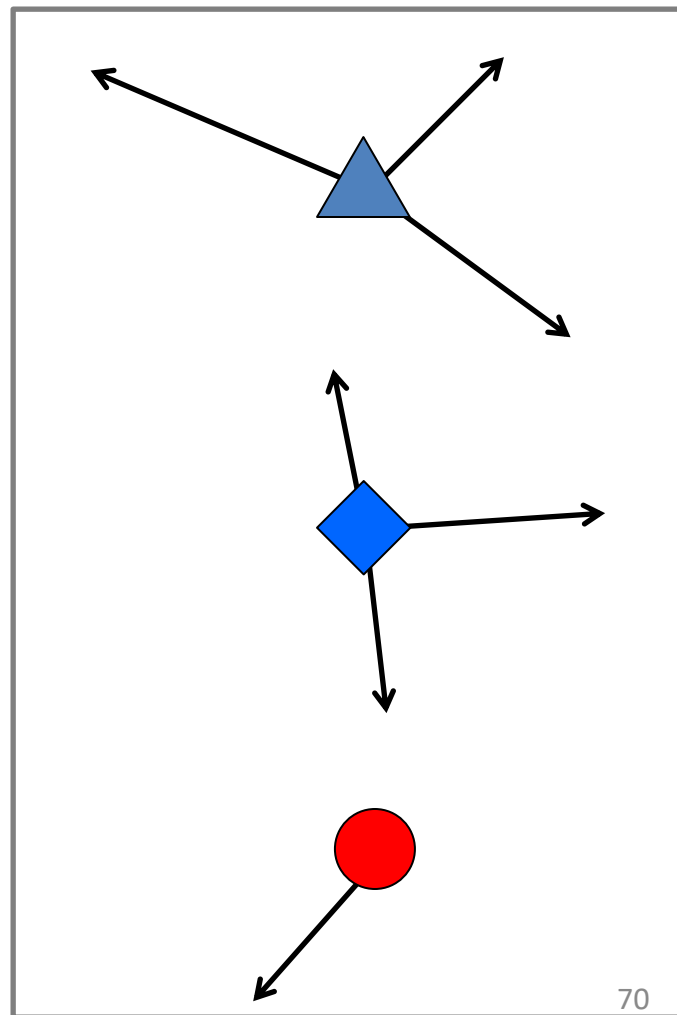
通用霍夫变换——检测

检测时，利用基于特征构建的模型
进行投票

Test image



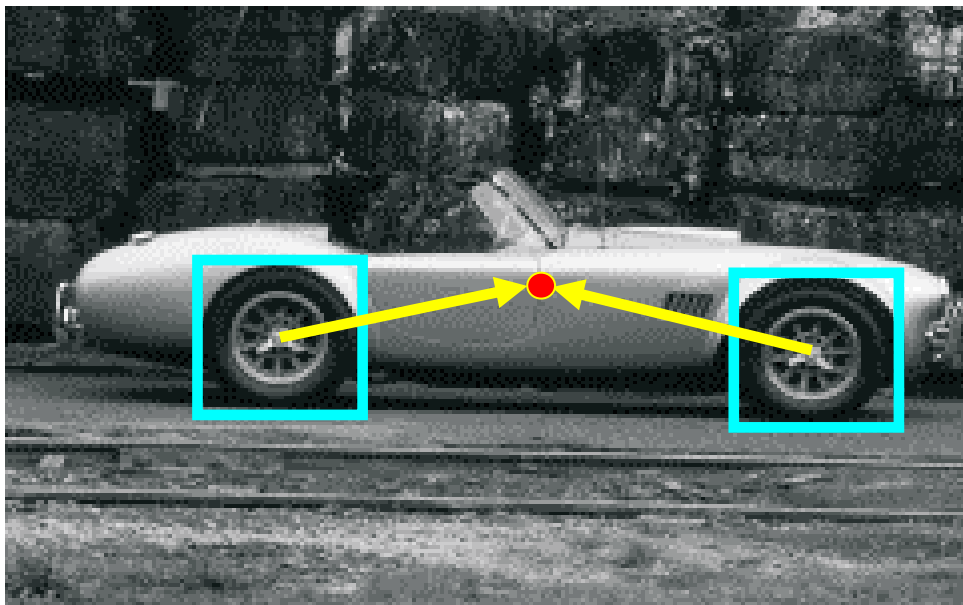
Model



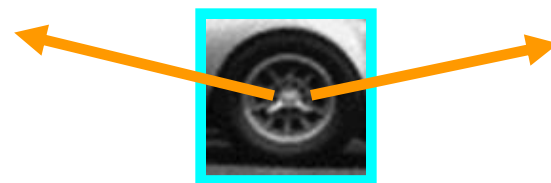


通用霍夫变换——检测

- 基于视觉字的投票



training image



visual codeword with
displacement vectors

B. Leibe, A. Leonardis, and B. Schiele, [Combined Object Categorization and Segmentation with an Implicit Shape Model](#), ECCV Workshop on Statistical Learning in Computer Vision 2004



通用霍夫变换——检测

- 基于视觉字的投票



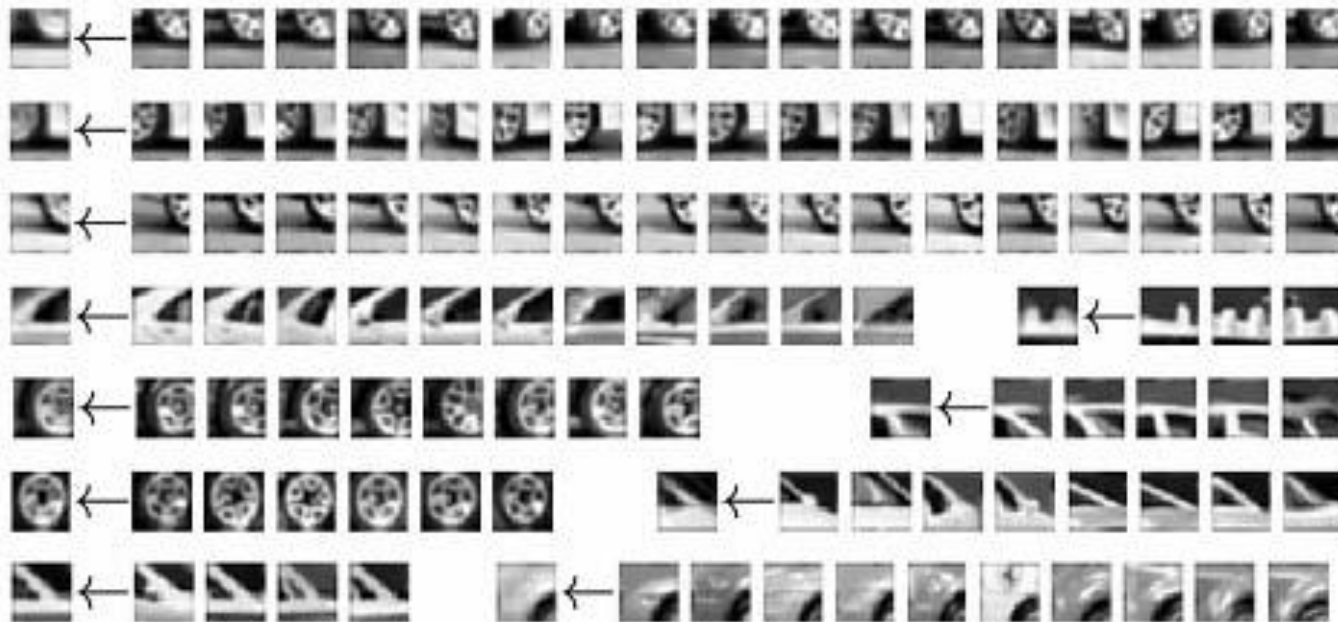
test image

B. Leibe, A. Leonardis, and B. Schiele, [Combined Object Categorization and Segmentation with an Implicit Shape Model](#), ECCV Workshop on Statistical Learning in Computer Vision 2004



Implicit shape models: Training

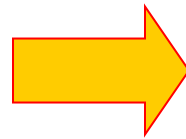
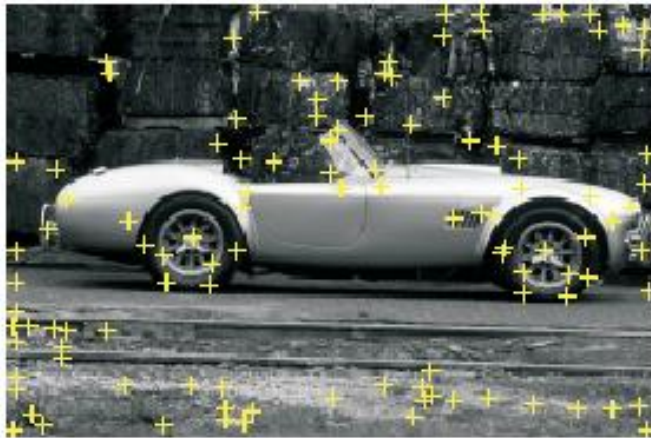
1. Build *codebook* of patches around extracted interest points using clustering





Implicit shape models: Training

1. Build *codebook* of patches around extracted interest points using clustering
2. Map the patch around each interest point to closest codebook entry



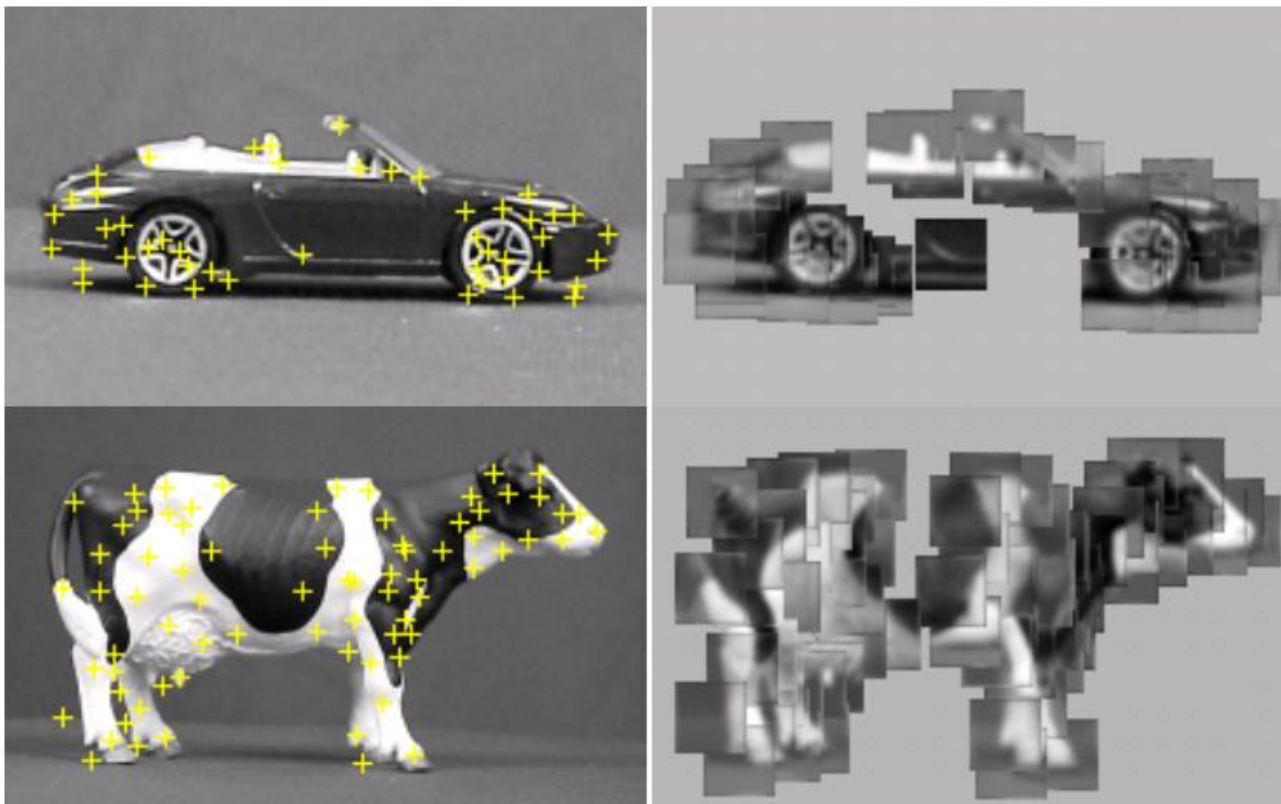
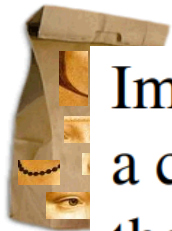
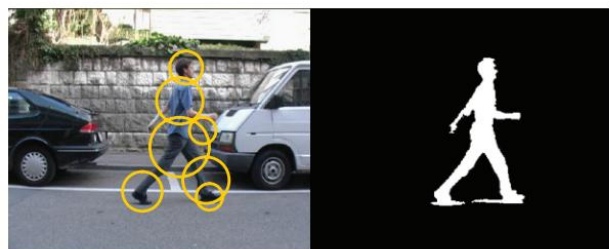


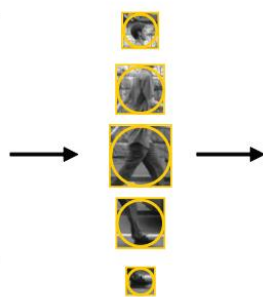
Fig. 1. *Local information used in the codebook generation process: (left) interest points; (right) features extracted around the interest points (visualized by the corresponding image patches). In most of our experiments, between 50 and 200 features are extracted per object.*



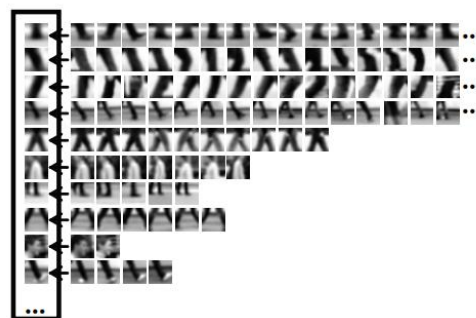
Implicit Shape Model $ISM(C) = (\mathcal{C}, P_C)$, which consists of a class-specific alphabet $\mathcal{C}$ (the *codebook*) of local appearances that are prototypical for the object category, and of a spatial probability distribution P_C which specifies where each codebook entry may be found on the object.



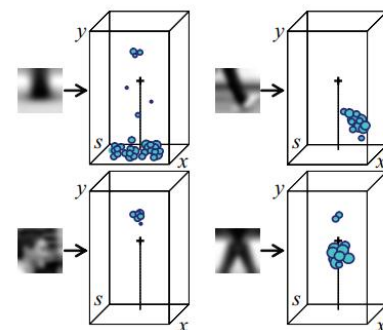
Training Images
(+Reference Segmentations)



Local
Features



Appearance Codebook
(Cluster Centers)

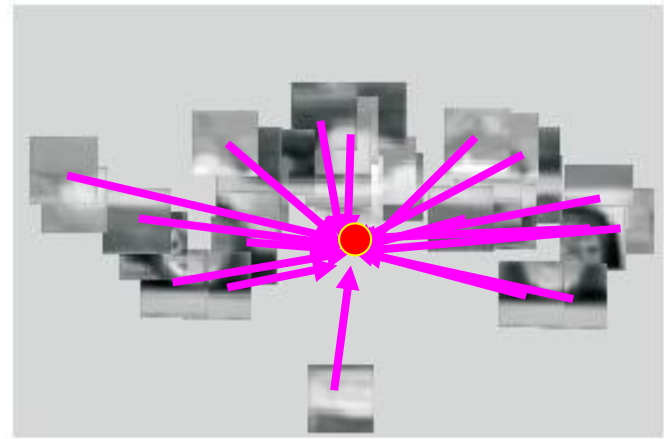
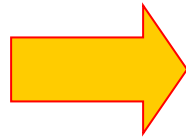
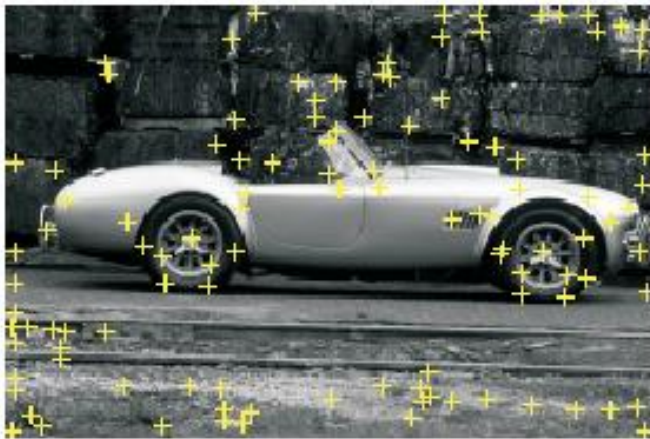


Spatial Occurrence Distributions
(non-parametric)



Implicit shape models: Training

1. Build *codebook* of patches around extracted interest points using clustering
2. Map the patch around each interest point to closest codebook entry
3. For each codebook entry, store all positions it was found, relative to object center



Algorithm 2 The training procedure.

// Create an appearance codebook $\mathcal{C}$.

$\mathcal{F} \leftarrow \emptyset$ *// Initialize the set of feature vectors $\mathcal{F}$*

for all training images **do**

 Apply the interest point detector.

for all interest regions $\ell_k = (\ell_x, \ell_y, \ell_s)$ with descriptors f_k **do**

$\mathcal{F} \leftarrow \mathcal{F} \cup f_k$

end for

end for

Cluster $\mathcal{F}$ with cut-off threshold t and keep cluster centers $\mathcal{C}$.

// Compute occurrences Occ .

for all codebook entries $\mathcal{C}_i$ **do**

$Occ[i] \leftarrow \emptyset$ *// Initialize occurrences for codebook entry $\mathcal{C}_i$*

end for

for all training images **do**

 Let (c_x, c_y) be the object center at a reference scale.

 Apply the interest point detector.

for all interest regions $\ell_k = (\ell_x, \ell_y, \ell_s)$ with descriptors f_k **do**

for all codebook entries $\mathcal{C}_i$ **do**

if $sim(\mathcal{C}_i, f_k) \geq t$ **then**

// Record an occurrence of codebook entry $\mathcal{C}_i$

$Occ[i] \leftarrow Occ[i] \cup (c_x - \ell_x, c_y - \ell_y, \ell_s)$

end if

end for

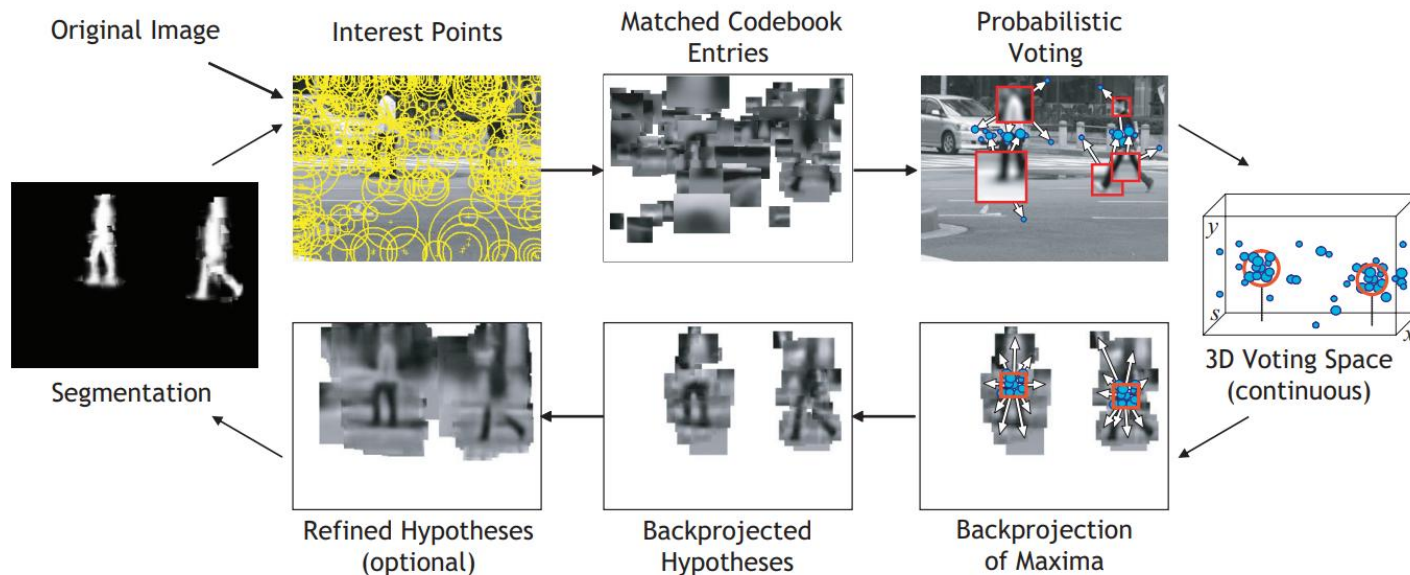
end for

end for



Recognition Approach

- 1) Probabilistic Hough Voting



The first term is the probabilistic Hough vote for an object position given its class label and the feature interpretation. The second term specifies a confidence that the codebook cluster is really matched on the target category as opposed to the background. This can be used to include negative examples in the training process. Finally, the third term reflects the quality of the match between image feature and codebook cluster.

$$\begin{aligned}
 p(o_n, x|f, \ell) &= \sum_i p(o_n, x|C_i, \ell)p(C_i|f). \\
 &= \sum_i p(x|o_n, C_i, \ell)p(o_n|C_i, \ell)p(C_i|f).
 \end{aligned}$$

Algorithm 3 The ISM vote generation algorithm.

// Initialize the set of probabilistic votes $\mathcal{V}$.

$\mathcal{V} \leftarrow \emptyset$

Apply the interest point detector to the test image.

for all interest regions $\ell_k = (\ell_x, \ell_y, \ell_s)$ with descriptors f_k **do**

// Initialize the set of matches $\mathcal{M}$

$\mathcal{M} \leftarrow \emptyset$

// Record all matches to the codebook

for all codebook entries C_i **do**

if $\text{sim}(f_k, C_i) \geq t$ **then**

$\mathcal{M} \leftarrow \mathcal{M} \cup (i, \ell_x, \ell_y, \ell_s)$ *// Record a match*

end if

end for

for all matching codebook entries C_i^* **do**

$p(C_i^* | f_k) \leftarrow \frac{1}{|\mathcal{M}|}$ *//Set the match weight*

end for

// Cast the votes

for all matches $(i, \ell_x, \ell_y, \ell_s) \in \mathcal{M}$ **do**

for all occurrences $\text{occ} \in \text{Occ}[i]$ of codebook entry C_i **do**

//Set the vote location

$x \leftarrow (\ell_x - \text{occ}_x \frac{\ell_s}{\text{occ}_s}, \ell_y - \text{occ}_y \frac{\ell_s}{\text{occ}_s}, \frac{\ell_s}{\text{occ}_s})$

//Set the occurrence weight

$p(o_n, x | C_i, \ell) \leftarrow \frac{1}{|\text{Occ}[i]|}$

// Cast a vote (x, w, occ, ℓ) for position x with weight w

$w \leftarrow p(o_n, x | C_i, \ell) p(C_i | f_k)$

$\mathcal{V} \leftarrow \mathcal{V} \cup (x, w, \text{occ}, \ell)$

end for

end for

end for





The end !



How do we evaluate object detection?



- • predictions
- • ground truth



How do we evaluate object detection?



- • predictions
- • ground truth

True positive:

- The overlap of the prediction with the ground truth is **MORE** than 0.5



How do we evaluate object detection?



- • predictions
- • ground truth

True positive:

False positive:

- The overlap of the prediction with the ground truth is **LESS** than 0.5



How do we evaluate object detection?



- • predictions
- • ground truth

True positive:

False positive:

False negative:

- The objects that our model doesn't find



How do we evaluate object detection?



- • predictions
- • ground truth

True positive:

False positive:

False negative:

- The objects that our model doesn't find

What is a **True Negative?**



	<u>Predicted 1</u>	<u>Predicted 0</u>
<u>True 1</u>	true positive	false negative
<u>True 0</u>	false positive	true negative





	<u>Predicted 1</u>	<u>Predicted 0</u>
<u>True 1</u>	true positive	false negative
<u>True 0</u>	false positive	true negative



	<u>Predicted 1</u>	<u>Predicted 0</u>
<u>True 1</u>	TP	FN
<u>True 0</u>	FP	TN



	<u>Predicted 1</u>	<u>Predicted 0</u>
<u>True 1</u>	true positive	false negative
<u>True 0</u>	false positive	true negative

=

	<u>Predicted 1</u>	<u>Predicted 0</u>
<u>True 1</u>	TP	FN
<u>True 0</u>	FP	TN

	<u>Predicted 1</u>	<u>Predicted 0</u>
<u>True 1</u>	hits	misses
<u>True 0</u>	false alarms	correct rejections



	<u>Predicted 1</u>	<u>Predicted 0</u>
<u>True 1</u>	true positive	false negative
<u>True 0</u>	false positive	true negative

=

	<u>Predicted 1</u>	<u>Predicted 0</u>
<u>True 1</u>	TP	FN
<u>True 0</u>	FP	TN

	<u>Predicted 1</u>	<u>Predicted 0</u>
<u>True 1</u>	hits	misses
<u>True 0</u>	false alarms	correct rejections

$$precision = \frac{TP}{TP + FP}$$

$$recall = \frac{TP}{TP + FN}$$



How do we evaluate object detection?



- • predictions
- • ground truth

True positive: 1

False positive: 2

False negative: 1

So what is the

- precision?

- recall?

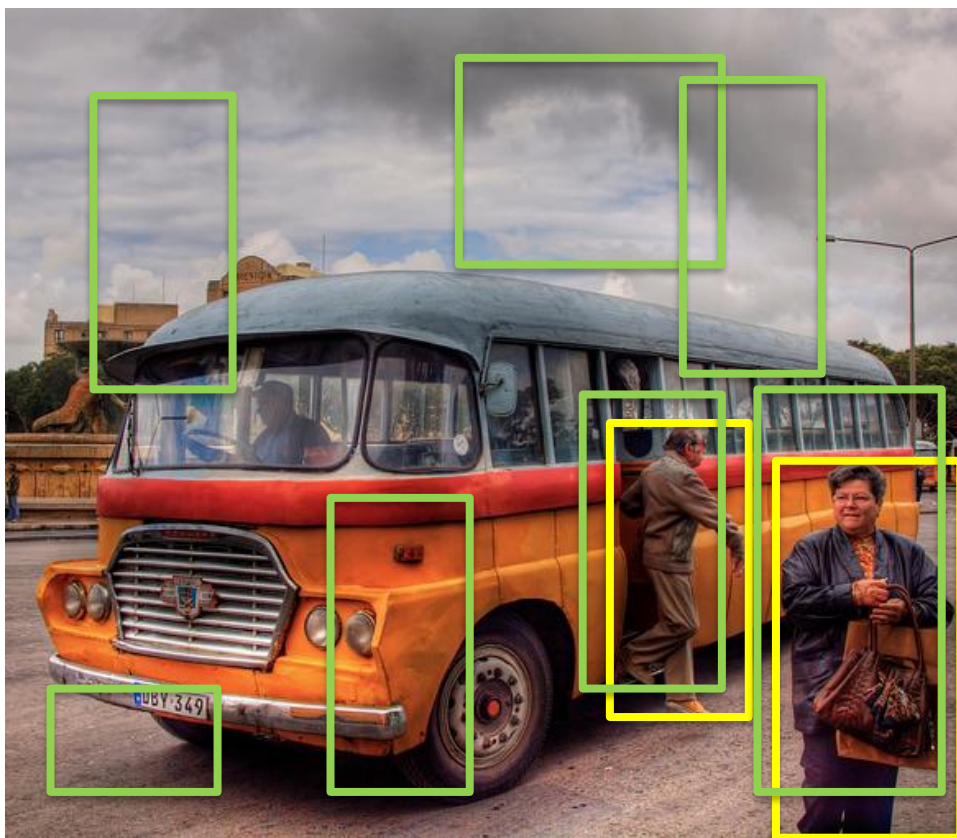


Precision versus recall

- Precision:
 - how many of the object detections are correct precision
- Recall:
 - how many of the ground truth objects can the model detect?



In reality, our model makes a lot of predictions with varying scores between 0 and 1



- • predictions
- • ground truth

Here are all the boxes that are predicted with $\text{score} > 0$.

This means that our

- Recall is perfect!
- But our precision is BAD!



In reality, our model makes a lot of predictions with varying scores between 0 and 1



- • predictions
- • ground truth

There are no boxes that are predicted with **score = 1**.

This means that our

- Precision is undefined!
- And our recall is BAD!



How do we evaluate object detection?



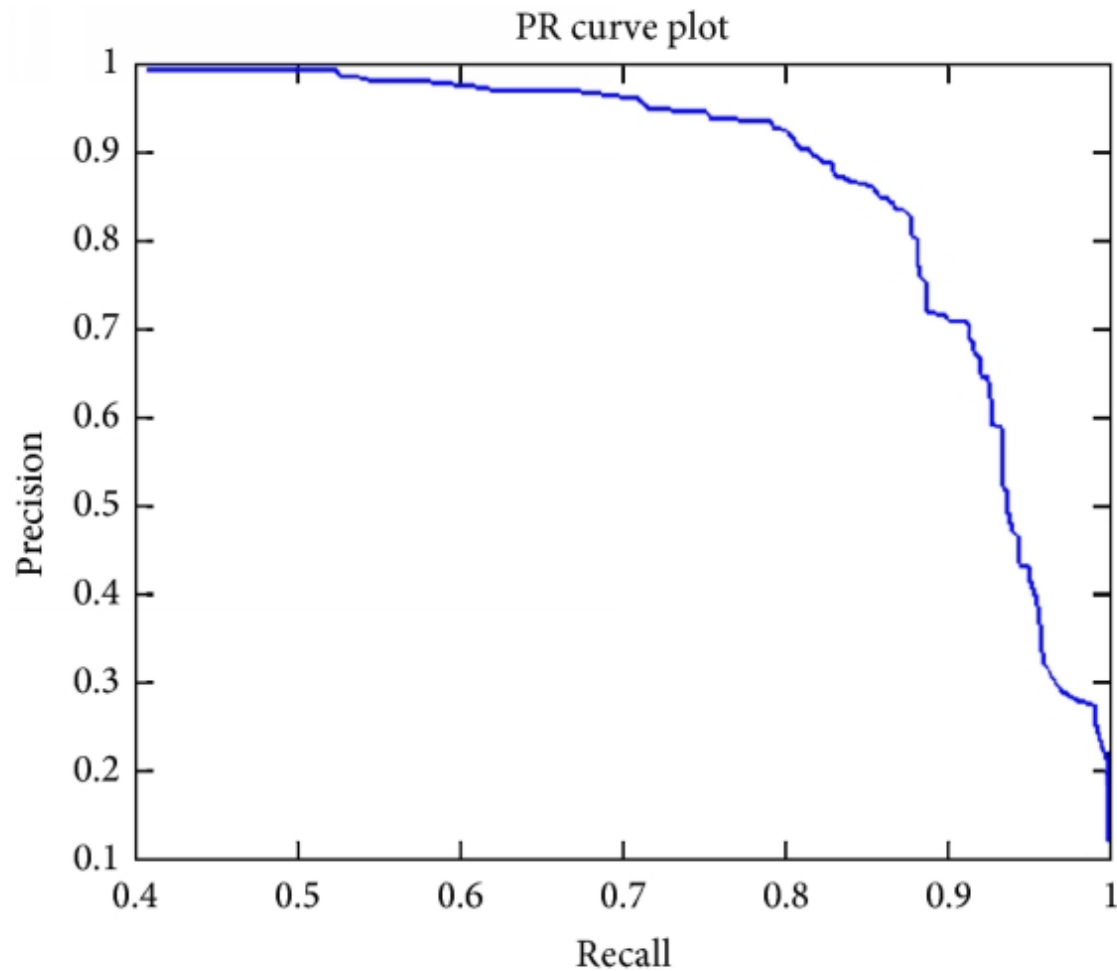
- • predictions
- • ground truth

Here are all the boxes
that are predicted with
score > 0.5

We are setting a
threshold of 0.5



Precision – recall curve (PR curve)





Which model is the best?

